
Paper Summaries

**Batches 01, 02 & 03 — Measurement Foundations,
Business Cycle Labour Flows & Firm Dynamics**

RDC Research Project

Labour Market Dynamics

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Summaries prepared in the role of a senior academic economist specialising in macroeconomics, labour economics, and applied econometrics. Each summary covers research question, contribution, methodology, results, limitations, policy implications, and research gaps across 19 structured sections.

Contents

1	Davis & Haltiwanger (1992)	8
1.1	Research Question	8
1.2	Motivation	8
1.3	Main Contribution	8
1.4	Relationship to the Literature	9
1.5	Theoretical Framework and Economic Mechanism	9
1.6	Data	10
1.7	Empirical Strategy	10
1.8	Identification Strategy	10
1.9	Main Results	11
1.10	Economic Magnitude and Interpretation	12
1.11	Mechanisms	12
1.12	Robustness Checks	12
1.13	Limitations	12
1.14	Policy Implications	13
1.15	Overall Assessment	13
1.16	Relevance to My Research	13
1.17	Potential Research Extensions	13
1.18	Research Gaps	14
1.19	Methodological Lessons	14
2	Hopenhayn (1992)	16
2.1	Research Question	16
2.2	Motivation	16
2.3	Main Contribution	16
2.4	Relationship to the Literature	17
2.5	Theoretical Framework and Economic Mechanism	17
2.6	Data	18
2.7	Empirical Strategy	18
2.8	Identification Strategy	18
2.9	Main Results	18

2.10	Economic Magnitude and Interpretation	19
2.11	Mechanisms	20
2.12	Robustness Checks	20
2.13	Limitations	20
2.14	Policy Implications	21
2.15	Overall Assessment	21
2.16	Relevance to My Research	21
2.17	Potential Research Extensions	21
2.18	Research Gaps	22
2.19	Methodological Lessons	22
3	Campbell & Fisher (2000)	24
3.1	Research Question	24
3.2	Motivation	24
3.3	Main Contribution	24
3.4	Relationship to the Literature	25
3.5	Theoretical Framework and Economic Mechanism	25
3.6	Data	26
3.7	Empirical Strategy	26
3.8	Identification Strategy	26
3.9	Main Results	27
3.10	Economic Magnitude and Interpretation	27
3.11	Mechanisms	27
3.12	Robustness Checks	28
3.13	Limitations	28
3.14	Policy Implications	28
3.15	Overall Assessment	29
3.16	Relevance to My Research	29
3.17	Potential Research Extensions	29
3.18	Research Gaps	29
3.19	Methodological Lessons	30
4	Davis & Haltiwanger (1999)	32
4.1	Research Question	32
4.2	Motivation	32
4.3	Main Contribution	32
4.4	Relationship to the Literature	33

4.5	Theoretical Framework and Economic Mechanism	33
4.6	Data	34
4.7	Empirical Strategy	35
4.8	Identification Strategy	35
4.9	Main Results	35
4.10	Economic Magnitude and Interpretation	36
4.11	Mechanisms	36
4.12	Robustness Checks	37
4.13	Limitations	37
4.14	Policy Implications	37
4.15	Overall Assessment	38
4.16	Relevance to My Research	38
4.17	Potential Research Extensions	38
4.18	Research Gaps	39
4.19	Methodological Lessons	39
5	Lazear & Spletzer (2012)	41
5.1	Research Question	41
5.2	Motivation	41
5.3	Main Contribution	41
5.4	Relationship to the Literature	42
5.5	Theoretical Framework and Economic Mechanism	42
5.6	Data	42
5.7	Empirical Strategy	43
5.8	Identification Strategy	43
5.9	Main Results	43
5.10	Economic Magnitude and Interpretation	44
5.11	Mechanisms	44
5.12	Robustness Checks	45
5.13	Limitations	45
5.14	Policy Implications	45
5.15	Overall Assessment	46
5.16	Relevance to My Research	46
5.17	Potential Research Extensions	46
5.18	Research Gaps	46
5.19	Methodological Lessons	47

6 Hyatt & McEntarfer (2012)	49
6.1 Research Question	49
6.2 Motivation	49
6.3 Main Contribution	49
6.4 Relationship to the Literature	50
6.5 Theoretical Framework and Economic Mechanism	50
6.6 Data	50
6.7 Empirical Strategy	50
6.8 Identification Strategy	51
6.9 Main Results	51
6.10 Economic Magnitude and Interpretation	51
6.11 Mechanisms	52
6.12 Robustness Checks	52
6.13 Limitations	52
6.14 Policy Implications	52
6.15 Overall Assessment	52
6.16 Relevance to My Research	53
6.17 Potential Research Extensions	53
6.18 Research Gaps	53
6.19 Methodological Lessons	53
7 Mueller (2017)	55
7.1 Research Question	55
7.2 Motivation	55
7.3 Main Contribution	55
7.4 Relationship to the Literature	56
7.5 Theoretical Framework and Economic Mechanism	56
7.6 Data	57
7.7 Empirical Strategy	57
7.8 Identification Strategy	57
7.9 Main Results	58
7.10 Economic Magnitude and Interpretation	58
7.11 Mechanisms	59
7.12 Robustness Checks	59
7.13 Limitations	59
7.14 Policy Implications	59

7.15 Overall Assessment	60
7.16 Relevance to My Research	60
7.17 Potential Research Extensions	60
7.18 Research Gaps	60
7.19 Methodological Lessons	60
8 Krusell, Mukoyama, Rogerson & Şahin (2017)	62
8.1 Research Question	62
8.2 Motivation	62
8.3 Main Contribution	62
8.4 Relationship to the Literature	63
8.5 Theoretical Framework and Economic Mechanism	63
8.6 Data	64
8.7 Empirical Strategy	64
8.8 Identification Strategy	64
8.9 Main Results	65
8.10 Economic Magnitude and Interpretation	65
8.11 Mechanisms	65
8.12 Robustness Checks	66
8.13 Limitations	66
8.14 Policy Implications	66
8.15 Overall Assessment	66
8.16 Relevance to My Research	67
8.17 Potential Research Extensions	67
8.18 Research Gaps	67
8.19 Methodological Lessons	67
9 Carrillo-Tudela & Visschers (2023)	69
9.1 Research Question	69
9.2 Motivation	69
9.3 Main Contribution	69
9.4 Relationship to the Literature	70
9.5 Theoretical Framework and Economic Mechanism	70
9.6 Data	71
9.7 Empirical Strategy	71
9.8 Identification Strategy	72
9.9 Main Results	72

9.10 Economic Magnitude and Interpretation	72
9.11 Mechanisms	73
9.12 Robustness Checks	73
9.13 Limitations	73
9.14 Policy Implications	74
9.15 Overall Assessment	74
9.16 Relevance to My Research	74
9.17 Potential Research Extensions	74
9.18 Research Gaps	74
9.19 Methodological Lessons	75
10 Asplund & Nocke (2006)	77
10.1 Research Question	77
10.2 Motivation	77
10.3 Main Contribution	77
10.4 Relationship to the Literature	78
10.5 Theoretical Framework and Economic Mechanism	78
10.6 Data	79
10.7 Empirical Strategy	79
10.8 Identification Strategy	80
10.9 Main Results	80
10.10 Economic Magnitude and Interpretation	80
10.11 Mechanisms	80
10.12 Robustness Checks	81
10.13 Limitations	81
10.14 Policy Implications	81
10.15 Overall Assessment	82
10.16 Relevance to My Research	82
10.17 Potential Research Extensions	82
10.18 Research Gaps	82
10.19 Methodological Lessons	82
11 Acemoglu, Akcigit, Alp, Bloom & Kerr (2018)	84
11.1 Research Question	84
11.2 Motivation	84
11.3 Main Contribution	84
11.4 Relationship to the Literature	85

11.5 Theoretical Framework and Economic Mechanism	85
11.6 Data	86
11.7 Empirical Strategy	86
11.8 Identification Strategy	87
11.9 Main Results	87
11.10Economic Magnitude and Interpretation	88
11.11Mechanisms	88
11.12Robustness Checks	88
11.13Limitations	89
11.14Policy Implications	89
11.15Overall Assessment	89
11.16Relevance to My Research	89
11.17Potential Research Extensions	89
11.18Research Gaps	90
11.19Methodological Lessons	90
12 Hopenhayn, Neira & Singhania (2022)	92
12.1 Research Question	92
12.2 Motivation	92
12.3 Main Contribution	92
12.4 Relationship to the Literature	93
12.5 Theoretical Framework and Economic Mechanism	93
12.6 Data	95
12.7 Empirical Strategy	95
12.8 Identification Strategy	95
12.9 Main Results	96
12.10Economic Magnitude and Interpretation	96
12.11Mechanisms	97
12.12Robustness Checks	97
12.13Limitations	97
12.14Policy Implications	98
12.15Overall Assessment	98
12.16Relevance to My Research	98
12.17Potential Research Extensions	98
12.18Research Gaps	98
12.19Methodological Lessons	99

Paper 1

Davis & Haltiwanger (1992)

Gross Job Creation, Gross Job Destruction, and Employment Reallocation

Authors: Steven J. Davis & John Haltiwanger

Journal: *Quarterly Journal of Economics* 107(3): 819–863

Year: 1992 **Field:** Labour economics; Macroeconomics

Classification: Empirical (descriptive measurement)

1.1 Research Question

What is the magnitude and structure of gross job creation and destruction in U.S. manufacturing, how does job reallocation relate to worker reallocation, and what accounts for simultaneous job creation and destruction within narrowly defined sectors of the economy?

1.2 Motivation

Prior to this paper, macroeconomics focused almost exclusively on *net* employment changes, treating aggregate employment as if it reflected homogeneous expansions and contractions across the economy. The large gross worker flows documented in the CPS literature (Clark–Summers, Abowd–Zellner) remained unexplained in structural terms. It was unclear whether large worker flows primarily reflected temporary layoffs and recalls across a fixed set of jobs, or whether a large portion was driven by the genuine destruction and creation of employment opportunities. The answer matters: if most worker mobility is driven by job reallocation, the character, timing, and welfare cost of unemployment are very different from a world where workers simply shuffle across stable jobs.

1.3 Main Contribution

The paper makes four major contributions.

1. **Measurement framework.** Constructs the DH gross job flow framework—the definitions of POS (job creation), NEG (job destruction), SUM (job reallocation), and the Davis–Haltiwanger symmetric growth rate measure—which became the field standard. U.S. manufacturing averaged 9.2% job creation and 11.3% job destruction per year (1972–1986).
2. **Job–worker reallocation link.** Demonstrates that 35–56% of all worker reallocation between employment states is directly driven by job reallocation: shifts in the distribution of employment opportunities across plants are a

dominant source of worker mobility.

3. **Persistence and concentration.** Job creation and destruction are largely *persistent* (one-year rates: 68% for creation, 81% for destruction) and *concentrated* (77% of destruction from plants shrinking by >20%), ruling out temporary layoff/recall as the dominant mechanism.
4. **Cross-sectional heterogeneity.** Documents systematic variation in job reallocation by plant age, size, ownership type, and region; shows that between-sector shifts account for essentially none ($\approx 12\%$ at the four-digit level) of excess job reallocation.

1.4 Relationship to the Literature

The paper sits at the intersection of three literatures: (i) gross worker flows (Clark–Summers 1979; Hall 1982; Blanchard–Diamond 1990); (ii) firm/plant-level employment dynamics (Jovanovic 1982; Evans 1987; Dunne, Roberts, Samuelson 1989); and (iii) macroeconomic business-cycle theory (Caballero 1990; Davis–Haltiwanger 1990). Among U.S. studies, only Leonard (1987) and Dunne, Roberts, Samuelson (1989) had previously measured gross job flows at the plant level, but with less frequency, shorter samples, and narrower coverage. The paper directly motivates Hopenhayn (1992) and much of the ensuing structural literature on firm dynamics.

1.5 Theoretical Framework and Economic Mechanism

The paper is primarily measurement-oriented. The conceptual apparatus rests on three mechanisms proposed in the literature to explain simultaneous job creation and destruction:

- **Passive learning and selection** (Jovanovic 1982): plants learn their efficiency type; those discovering unfavourable types exit while survivors grow.
- **Active learning / vintage effects** (Ericson–Pakes 1989): stochastic investment outcomes generate ongoing idiosyncratic uncertainty, sustaining perpetual entry and exit.
- **Idiosyncratic cost/demand disturbances** (Hopenhayn 1989; Davis–Haltiwanger 1990): even homogeneous plants within sectors face local idiosyncratic shocks, driving simultaneous expansion and contraction.

Measurement Definitions

The growth rate of establishment e at time t is defined as:

$$g_{et} = \frac{E_{et} - E_{e,t-1}}{\frac{1}{2}(E_{et} + E_{e,t-1})}, \quad (1.1)$$

which lies in $[-2, 2]$ and symmetrically handles births ($g = 2$) and deaths ($g = -2$). Gross job creation and destruction rates for sector s at time t are:

$$\text{POS}_{st} = \sum_{\substack{e \in \mathcal{E}_{st} \\ g_{et} > 0}} \frac{x_{et}}{X_{st}} g_{et}, \quad (1.2)$$

$$\text{NEG}_{st} = \sum_{\substack{e \in \mathcal{E}_{st} \\ g_{et} < 0}} \frac{x_{et}}{X_{st}} |g_{et}|, \quad (1.3)$$

where $x_{et} = \frac{1}{2}(E_{et} + E_{e,t-1})$ is the size metric and X_{st} is the sector-level analogue. The job reallocation rate is $\text{SUM}_{st} = \text{POS}_{st} + \text{NEG}_{st}$. The required worker reallocation is bounded by $\text{MAX}_{st} = \max\{\text{POS}_{st}, \text{NEG}_{st}\}$ (lower bound) and SUM_{st} (upper bound).

1.6 Data

- **Dataset:** Longitudinal Research Datafile (LRD), U.S. Census Bureau.
- **Coverage:** All manufacturing establishments with 5+ employees sampled in the Annual Survey of Manufactures (ASM).
- **Sample:** $\approx 860,000$ annual observations on $\approx 160,000$ establishments; 11 years (1972–1986, excluding 1974, 1979, 1984 due to ASM panel break-years).
- **Strengths:** Probability-based sample; births incorporated into ongoing panels; careful distinction between ownership transfers and actual births/deaths; certainty stratum covers $\approx$ two-thirds of manufacturing employment (≥ 250 employees).
- **Limitation:** Five-year panel structure makes non-certainty (small) establishments harder to track across panel boundaries; point-in-time (March) employment understates true intra-year flows.

1.7 Empirical Strategy

Pure descriptive measurement. The strategy involves:

1. Computing annual POS, NEG, SUM, MAX at the establishment level and aggregating using employment shares as weights.
2. Tabulating *persistence*: what fraction of jobs created (destroyed) in year t remain created (destroyed) one and two years later.
3. Cross-tabulating gross flow rates by industry, size class, age group, ownership type, and geographic region.
4. Bounding required worker reallocation (SUM upper bound; MAX lower bound) and comparing against CPS-based total worker reallocation.

1.8 Identification Strategy

This is a descriptive paper with no causal identification. The approach is measurement and decomposition. For the passive-learning exercise, the authors use an indirect decomposition: plants older than some age T^* are assumed to have fully resolved initial uncertainty, so the excess reallocation of younger plants relative to

this benchmark is attributed to learning. This requires two identifying assumptions: (i) within-industry, long-run job reallocation rates are stable across age cohorts for old plants; (ii) excess reallocation of young plants is not driven by other factors that diminish mechanically with age.

1.9 Main Results

Core Measurement (Table I)

Statistic	Value	Period
Average annual job creation rate (POS)	9.2%	1972–1986
Average annual job destruction rate (NEG)	11.3%	1972–1986
Average job reallocation rate (SUM)	20.5%	1972–1986
Range of job reallocation rate	17.3–23.3%	
$\rho(\text{POS}, \text{NEG})$	−0.864	
$\rho(\text{NET}, \text{SUM})$	−0.565	

Cross-industry variation (Table II)

Job reallocation ranges from 14.0% (Tobacco) to 28.8% (Lumber) across 2-digit industries. Every industry exhibits simultaneous creation and destruction; excess reallocation ranges from 9.8% to 20.6%.

Persistence (Table III)

	Job Creation	Job Destruction
One-year persistence (mean)	68%	81%
Two-year persistence (mean)	50%	73%

Plant characteristics (Table IV)

- **Age:** reallocation falls from 48% at age 1 to 16% at 15+ years—the sharpest gradient in the data.
- **Size:** reallocation falls from 30% (1–99 employees) to 14% (1000+ employees).
- **Ownership:** single-unit plants 28%; multi-unit 18%.

Job–worker reallocation link

Total worker reallocation $\approx 36.8\%$ of employment annually (CPS). Job reallocation accounts for 35–56% of total worker reallocation.

1.10 Economic Magnitude and Interpretation

The magnitude is striking. In an average year, employment positions equivalent to one-fifth of the manufacturing workforce are either newly created or newly destroyed. Even in boom years, gross job destruction exceeds 6%. The persistence results imply that most job creation and destruction represents real, durable changes—not temporary layoff/recall cycles. The concentration result (77% of destruction from plants shrinking >20%) means most job losses cannot be absorbed by normal attrition and require actual layoffs. These findings establish that standard representative-agent macro models, which focus entirely on net employment, ignore enormous amounts of economically significant labour market activity.

1.11 Mechanisms

The paper evaluates four potential explanations:

1. **Between-sector employment shifts:** Even at the 4-digit industry level (450 industries), between-sector shifts account for only $\approx 12\%$ of excess job reallocation. Sectoral-shock explanations are definitively ruled out.
2. **Passive learning:** Accounts for only 11–13% of *total* reallocation, but explains one-third to one-half of the *variation* in reallocation rates across age, size, and ownership groups.
3. **Plant-level idiosyncratic disturbances:** By elimination, the dominant mechanism must be ongoing plant-level idiosyncratic shocks orthogonal across plants even within the same 4-digit industry.
4. **Cyclical variation:** The countercyclical pattern reflects time variation in the *magnitude* of idiosyncratic shocks, not compositional shifts. Patterns are most pronounced in older, larger, multi-unit plants.

1.12 Robustness Checks

- Consistency of gross flow estimates with LRD-period BLS Business Employment Dynamics data.
- Consistency of CPS-based gross worker flow adjustments using both Abowd–Zellner and Poterba–Summers error matrices.
- Results hold across multiple base years for persistence calculations.
- Cross-industry variation results replicated across individual years.

Key limitation acknowledged: The LRD excludes establishments with <5 employees. Leonard (1987) finds nonmanufacturing reallocation is $\approx 28\%$ higher than manufacturing, suggesting economy-wide reallocation substantially exceeds the manufacturing estimates.

1.13 Limitations

1. **Sector coverage:** Confined to U.S. manufacturing (declining over the period). Service sectors, with different dynamics, are excluded.

2. **Annual frequency:** March-to-March data miss intra-year job flows; reported annual flows understate true quarterly or monthly flows.
3. **Point-in-time employment:** Measures lower bounds on true gross flows.
4. **No causal identification:** Documents regularities but cannot explain *why* reallocation is countercyclical.
5. **Learning decomposition is fragile:** Requires identifying assumptions about when reallocation stabilises at its long-run level.

1.14 Policy Implications

- High reallocation rates imply unemployment is typically a structural transition phenomenon; job-search assistance and training are more relevant than aggregate demand stimulus.
- Countercyclical reallocation supports “cleansing recession” arguments against bailouts that preserve misallocated resources.
- Firing-cost regulations would reduce both creation and reallocation, potentially lowering long-run productivity.
- Large role of young and small plants in total reallocation suggests that entry costs (regulations, administrative burdens) have first-order effects on allocative efficiency.

1.15 Overall Assessment

This is a landmark empirical paper that created a field. The DH gross job flow framework is now used worldwide; the LRD-based estimates have been replicated and extended to dozens of countries and sectors. The paper’s weaknesses are largely a function of when it was written: limited coverage of small plants, manufacturing-only scope, and no causal identification. The paper does not attempt to distinguish whether countercyclical reallocation is efficient (“cleansing”) or inefficient (“sully”), a distinction that would motivate a decade of subsequent theoretical work.

Quality: Exceptional. Required reading for anyone working on labour dynamics, firm dynamics, or macroeconomic fluctuations.

1.16 Relevance to My Research

Directly foundational for RDC-based research on job flows in Canada. The methodology (plant-level longitudinal data, decomposition of gross flows by plant characteristics) is exactly the approach applicable to LEAP or LEED data. The key measurements would be directly replicable with Canadian data and would reveal whether the magnitudes found in U.S. manufacturing generalise to the Canadian context—including resource-sector-heavy Western provinces.

1.17 Potential Research Extensions

- Replicate the DH framework for Canadian manufacturing and services using LEAP/LEED; test whether Canadian job reallocation is higher or lower than U.S., particularly around resource price cycles.

- Compare job reallocation dynamics before and after major trade liberalisations (FTA 1989, NAFTA 1994, CETA).
- Examine whether countercyclical job reallocation intensified during the 2008–09 recession and COVID-19 in Canada.

1.18 Research Gaps

- **Causal mechanism:** Does countercyclical reallocation represent efficient cleansing or inefficient disruption? The paper is silent on this welfare question.
- **Worker outcomes:** Job destruction rates tell us about positions, not about workers. Earnings losses, nonemployment duration, and occupational switching are entirely outside the scope.
- **Service sector:** U.S. nonmanufacturing dynamics at the plant level remain underexplored.

1.19 Methodological Lessons

1. The DH growth rate measure $g_{et} = \Delta E / \text{average}(E)$ is superior to the log-difference because it handles births and deaths symmetrically within a unified framework.
2. Persistence measures (fraction of newly created/destroyed jobs still present one or two years later) are essential for distinguishing transitory from permanent employment changes.
3. Bounding worker reallocation from above (SUM) and below (MAX) is a simple and powerful diagnostic for the connection between job and worker flows.
4. Weighting establishment-level growth rates by employment share (rather than equally) is critical for ensuring aggregate implications reflect the behaviour of large employers.

Five-Sentence Summary

Davis and Haltiwanger (1992) use the U.S. Census Bureau’s Longitudinal Research Datafile to document, for the first time comprehensively, that gross job creation and destruction in U.S. manufacturing averaged 9.2% and 11.3% of employment per year over 1972–1986—flows an order of magnitude larger than net employment changes. They show that job reallocation is persistent (68–81% one-year persistence), concentrated (77% of destruction from plants shrinking >20%), and accounts for 35–56% of all worker reallocation, establishing that shifts in the distribution of employment opportunities are a primary driver of worker mobility. Cross-sectional analysis reveals that job reallocation rates decline sharply with plant age (48% at age 1; 16% at 15+ years) and size, and that virtually none of excess reallocation can be attributed to between-sector employment shifts even at the 4-digit industry level. The countercyclical pattern of job reallocation is driven by time variation in the magnitude of idiosyncratic plant-level shocks, primarily at older and larger establishments. The paper establishes the conceptual and measurement framework—the DH gross job flow decomposition—that has become the foundation for all subsequent empirical

and theoretical work on firm dynamics, labour market flows, and macroeconomic reallocation.

Quick Reference

Key contribution	First comprehensive plant-level measurement of gross job flows; establishes that job reallocation is large, persistent, and drives a substantial fraction of worker mobility.
Key weakness	Descriptive only; confined to manufacturing; annual frequency understates true flow magnitudes; no causal identification of the countercyclical reallocation mechanism.
Most interesting gap	What happens to workers displaced by job destruction—the paper is silent on earnings losses, nonemployment duration, and occupational switching.
Publishable extension	Replicate the DH framework for Canadian data (LEAP/LEED) across the full economy, with particular attention to resource-sector dynamics and commodity price cycles on job reallocation patterns.

Paper 2

Hopenhayn (1992)

Entry, Exit, and Firm Dynamics in Long Run Equilibrium

Author: Hugo A. Hopenhayn

Journal: *Econometrica* 60(5): 1127–1150

Year: 1992 **Field:** Applied microeconomics; Industrial organisation; Macroeconomics

Classification: Theoretical

2.1 Research Question

What determines the rates of firm entry, exit, and the stationary distribution of firm size and profits in a competitive industry subject to idiosyncratic productivity shocks, and how do changes in industry parameters (entry costs, fixed costs, demand) affect these equilibrium outcomes?

2.2 Motivation

By the late 1980s, empirical work (Davis–Haltiwanger 1991; Dunne, Roberts, Samuelson 1989) had documented two striking facts: (i) firm-specific idiosyncratic uncertainty dominates firm-size dynamics; and (ii) industry-level entry and exit rates are highly correlated, with large, persistent cross-industry differences. Existing theoretical models either lacked firm heterogeneity, lacked entry/exit in steady state, or were computationally intractable dynamical systems. Hopenhayn builds a tractable equilibrium model that generates ongoing entry and exit in the steady state, accommodates idiosyncratic productivity shocks, admits closed-form comparative statics, and can be connected to the growing empirical literature on firm heterogeneity.

2.3 Main Contribution

1. **Stationary equilibrium concept:** Extends long-run competitive industry equilibrium to a setting with idiosyncratic shocks, firm heterogeneity, and simultaneous entry and exit in steady state. Proves existence and uniqueness under standard conditions (Theorems 2, 3, 4).
2. **Comparative statics:** Derives clean predictions for how entry costs, fixed costs, and demand changes affect the exit threshold, firm turnover rate, and the stationary size distribution—including the counterintuitive result that higher entry costs *protect incumbents* and *reduce* turnover.
3. **Lifecycle implications:** Shows that under monotone conditional dominance

(mcd) ordering of the Markov transition, the distribution of firms' productivity shocks is stochastically increasing in cohort age—consistent with declining hazard rates, larger average size, and higher average profits for older cohorts (Propositions 3–5, 7–9).

2.4 Relationship to the Literature

The paper extends and synthesises three strands:

- **Jovanovic (1982)**: passive learning model has no stationary equilibrium with entry/exit (the stopping time has positive probability at infinity). Hopenhayn's recurrence assumption (A.4) rules this out.
- **Lucas–Prescott (1971)**: competitive equilibrium theory but with homogeneous firms and no entry/exit.
- **Ericson–Pakes (1989)**: rich active-learning model but computationally intractable for analytical work.

The paper directly motivates Hopenhayn–Rogerson (1993) on firing costs, Melitz (2003) on trade with heterogeneous firms, Asplund–Nocke (2006) on imperfect competition, and Acemoglu et al. (2018) on innovation and reallocation.

2.5 Theoretical Framework and Economic Mechanism

Industry Structure

A continuum of firms produce a homogeneous good. Output market is given by inverse demand $D(Q)$; input market (labour) by supply schedule $W(N)$. Firms face a fixed cost c_f per period and entry cost c_e (sunk). Individual firm output is $q = f(\varphi, n)$, where $\varphi \in [0, 1]$ is an idiosyncratic productivity shock evolving as a Markov process $F(\varphi'|\varphi)$ satisfying first-order stochastic dominance in φ (Assumption A.3(b)).

Firm's Bellman Equation

The value of an incumbent firm with productivity φ in a market with distribution μ of active firms satisfies:

$$v(\varphi, \mu) = \pi(\varphi, \mu) + \max\left\{0, \beta \int v(\varphi', \mu) F(d\varphi'|\varphi)\right\}, \quad (2.1)$$

where $\pi(\varphi, \mu)$ is current-period profit (net of fixed cost c_f). Because v is strictly increasing in φ (Proposition 3(i)), the optimal exit rule is a reservation rule: exit if and only if $\varphi < x^*$.

Equilibrium Conditions

Exit condition:

$$\int v(\varphi', \mu^*) F(d\varphi'|x^*) = 0. \quad (2.2)$$

Free entry condition:

$$\int v(\varphi, \mu^*) \nu(d\varphi) = c_e, \quad (2.3)$$

where ν is the initial distribution from which entrants draw their first productivity shock.

Invariant measure:

$$\mu^* = P_x \mu^* + M^* \nu, \quad (2.4)$$

where P_x is the transition operator conditional on survival ($\varphi \geq x^*$) and M^* is the equilibrium mass of entrants. The existence proof constructs two curves $M_1(x)$ (exit-consistent entry mass) and $M_2(x)$ (entry-consistent entry mass) and establishes their intersection corresponds to a stationary equilibrium with positive entry and exit (Theorem 3).

Lifecycle Mechanism

As cohorts age, the worst-performing firms ($\varphi < x^*$) are continuously removed by the exit rule. Under the mcd ordering of F , the distribution μ_t of cohort- t firms is stochastically increasing over time (Proposition 4), delivering declining hazard rates, rising average size, and rising average profits—all consistent with the empirical record.

2.6 Data

No empirical data. The paper is purely theoretical. Empirical facts motivating the framework (approximately one-third of manufacturing jobs and over 40% of firms disappear over five-year periods) are drawn from Davis–Haltiwanger (1991) and Dunne, Roberts, Samuelson (1989). A numerical example in Section 5 illustrates the comparative statics for the effect of fixed costs on the exit threshold.

2.7 Empirical Strategy

Not applicable. The model’s cross-sectional implications (declining hazard rates with age, increasing average size with age, regression to the mean) are compared informally to empirical evidence from Dunne, Roberts, and Samuelson. No formal estimation or hypothesis testing is performed.

2.8 Identification Strategy

Not applicable in a causal sense. The paper’s key analytical tool is the algorithmic existence proof: showing that x^* and M^* are solutions to the $M_1(x)$ – $M_2(x)$ crossing, and that this crossing is unique under Condition U.1 (price-taking in input markets) or U.2 (profit function separable in φ and prices). The proof is constructive and provides a computational roadmap for numerical implementation.

2.9 Main Results

Existence and Uniqueness

- **Theorem 2:** A stationary competitive equilibrium always exists.

- **Theorem 3:** A stationary equilibrium with positive entry and exit exists if and only if $c_e < c^*$ for some threshold c^* .
- **Theorem 4:** Under Condition U.1 or U.2, the equilibrium is unique.

Lifecycle Properties

Under the mcd ordering of F :

- The age-cohort distributions $\{\mu_t\}$ are stochastically increasing in t (Proposition 4).
- Hazard rates are declining in both age and current productivity (Proposition 5).
- Conditional on survival, older and larger firms have higher expected productivity, profits, and value.

Comparative Statics

Parameter change	Effect on x^*	Effect on turnover rate	Condition
$\uparrow c_e$	$\downarrow$ (less selection)	$\downarrow$	Always
$\uparrow c_f$	$\uparrow$ (more selection)	$\uparrow$	Under U.2; can reverse under U.1
Mean-preserving spread of ν	$\uparrow$	$\uparrow$	Under convexity of v
$\uparrow$ Demand (elastic supply)	No effect	No effect	Under U.1

Profit and Value Bounds

Proposition 7: Average industry profit satisfies $\bar{\pi}(\mu^*) \geq \rho c_e$, where $\rho = M^*/\mu^*(S)$ is the turnover rate. Turnover measures the minimum required average profit to sustain the entry incentive.

Proposition 8: Average firm value $\bar{V}(\mu^*) = \bar{\pi}(\mu^*)/(1 - \beta) - \beta \rho c_e/(1 - \beta)$, providing a formula for measuring implicit sunk entry costs from observed value and profit data.

2.10 Economic Magnitude and Interpretation

The paper is analytical; magnitudes are structural rather than calibrated. The key economic insight about magnitudes is the profit lower bound: an industry with a 20% annual turnover rate ($\rho = 0.2$) and entry cost equal to one year's revenue requires average annual profits of at least 20% of entry cost to sustain equilibrium entry. The numerical example in Section 5 illustrates that the effect of higher fixed costs on the exit threshold can *reverse sign* depending on whether the technology

features sufficiently high elasticity of profits with respect to productivity—a subtle but important caveat on the comparative statics.

2.11 Mechanisms

The central mechanism is **selection through exit**: the industry is in a constant process of creative destruction. Each period, the worst-performing firms ($\varphi < x^*$) exit, their resources are freed, and new entrants draw from ν . Three secondary mechanisms are worth noting:

- **Option value of staying:** An incumbent with φ slightly below x^* may remain if the continuation value from potential future productivity improvement exceeds current negative profit. This is why $x^* < \varphi_{\max}$ and why even slightly unprofitable firms survive temporarily.
- **Selection amplification by age:** As cohorts age, selection removes the worst performers, leaving an increasingly productive and stable surviving set—consistent with empirically documented declining hazard rates.
- **Competition via entry and prices:** Higher entry rates drive down prices, raising the exit threshold and accelerating exit of less productive incumbents. This general equilibrium price mechanism is absent from partial equilibrium analyses.

2.12 Robustness Checks

As a theoretical paper, robustness is assessed through:

- Proof of existence under broad conditions (Assumptions A.1–A.5 only).
- Sufficient conditions for uniqueness (U.1 or U.2, both widely satisfied by standard functional forms including CES and Cobb–Douglas).
- A numerical example demonstrating that the fixed-cost comparative static can reverse under U.1, providing an important caveat.
- Extension to multidimensional state spaces (CES production with multiple productivity parameters) shown to satisfy U.2.

2.13 Limitations

1. **No dynamics:** Focuses entirely on stationary equilibria. Transitional dynamics—how the industry responds to a parameter change before reaching the new stationary state—are not analysed.
2. **Competitive markets only:** Extension to imperfect competition requires Asplund–Nocke (2006); price-competition effects are absent.
3. **No aggregate shocks:** All aggregate variables (price, output, employment) are deterministic in the stationary state. There are no business cycles.
4. **Passive productivity process:** φ evolves exogenously. Firms cannot invest to change their position in the productivity distribution (handled by Ericson–Pakes and Acemoglu et al.).

5. **No worker side:** Labour is supplied elastically at price $W(N)$. Labour market frictions, search, or matching are outside the model.

2.14 Policy Implications

- **Entry barriers reduce turnover and protect incumbents:** Higher c_e reduces x^* , enabling less productive firms to survive. Entry barriers impose a double welfare cost: fewer entrants and a less efficient incumbent stock.
- **Fixed costs and selection:** Higher fixed production costs raise x^* , increasing average productivity by forcing exit of less productive firms. This provides a theoretical basis for why fixed-cost regulation can have ambiguous welfare effects.
- **Firing cost policies:** The paper (final remarks) notes that a version of the model is used in Hopenhayn–Rogerson (1993) to study firing costs. By raising effective fixed costs and reducing selection, firing costs reduce long-run productivity.

2.15 Overall Assessment

Hopenhayn (1992) is one of the most important theory papers in industrial organisation and macroeconomics of the last four decades. Its significance lies not in the complexity of the model—which is deliberately parsimonious—but in the analytical tractability it achieves while incorporating heterogeneity, entry, and exit simultaneously. The existence–uniqueness results, comparative statics, and lifecycle propositions have all become standard ingredients in quantitative macro models. Direct descendants include Melitz (2003), Hopenhayn–Rogerson (1993), and Acemoglu et al. (2018). The paper’s limitations are largely features: by keeping the model simple, Hopenhayn made it usable.

Quality: Essential. A must-read for anyone working on firm dynamics, industrial policy, or models with heterogeneous agents.

2.16 Relevance to My Research

Directly relevant as the theoretical foundation for any RDC project on firm entry, exit, and dynamics. The comparative statics on entry costs and fixed costs provide testable predictions for Canadian data: industries with lower entry barriers should show higher turnover rates and higher productivity distributions. The profit bound $\bar{\pi} \geq \rho c_e$ provides a formula evaluable with Canadian firm-level data (T2 corporate tax records).

2.17 Potential Research Extensions

- Estimate the model structurally using Canadian firm-level data to recover c_e , c_f , and the Markov process $F(\varphi'|\varphi)$.
- Exploit provincial variation in effective entry costs (regulatory burden indices) to test the comparative statics predictions on firm turnover and productivity distributions.
- Embed in a GE model with labour market search to study the interaction

between firm exit rates and unemployment duration.

2.18 Research Gaps

- **Transitional dynamics:** Policy reform evaluations require knowing the transition path, which the stationary analysis cannot provide.
- **Worker heterogeneity:** Workers are homogeneous; a natural extension would allow differential matching with firm types, generating wage variation driven by firm productivity heterogeneity.
- **Financial constraints:** Introducing credit constraints could generate amplification—firms below the exit threshold that would survive absent the constraint, but cannot access external finance.

2.19 Methodological Lessons

1. The stationary equilibrium concept—requiring that the invariant measure μ^* equals the measure generated by entry/exit rules and entry distribution—is the right way to think about long-run industry equilibria with heterogeneous firms.
2. The algorithmic existence proof (constructing $M_1(x)$ and $M_2(x)$ curves) provides a computational roadmap for numerical implementation— a rare feature of analytical theory papers.
3. The mcd (monotone conditional dominance) ordering is stronger than FOSD but weaker than likelihood ratio ordering, and survives the truncation from the exit threshold—a useful technical tool for dynamic selection models.

Five-Sentence Summary

Hopenhayn (1992) develops a tractable dynamic stochastic model of a competitive industry in which firms face idiosyncratic productivity shocks and choose optimally when to exit, with free entry driving expected profits to the entry cost. The concept of stationary competitive equilibrium—a steady-state where entry and exit rates are equal and the distribution of active firms is time-invariant—is introduced and shown to exist and be unique under broad conditions. Comparative statics demonstrate that higher entry costs reduce turnover by protecting less productive incumbents, while higher fixed costs (under separability) raise the exit threshold and improve average firm productivity through selection. Under monotone conditional dominance of the productivity process, the model implies that older cohorts of firms are stochastically larger, have lower hazard rates, and earn higher average profits—all consistent with the empirical evidence from Dunne, Roberts, and Samuelson. The paper provides the canonical theoretical framework for all subsequent quantitative work on firm dynamics, entry, exit, and the long-run effects of industrial policy on productivity and resource allocation.

Quick Reference

Key contribution	Tractable, analytically characterised stationary equilibrium with simultaneous entry and exit driven by idiosyncratic productivity; clean comparative statics on entry/fixed costs and demand.
Key weakness	No aggregate shocks (no business cycle); no transitional dynamics; perfectly competitive markets only; no investment in productivity.
Most interesting gap	Transitional dynamics after policy changes—a reform that reduces entry costs generates extended transitional dynamics that the stationary analysis cannot characterise.
Publishable extension	Structural estimation of the Hopenhayn model using Canadian T2 tax data, exploiting variation in effective provincial entry costs (regulatory burden indices) to identify the entry cost parameter and test comparative statics predictions on firm turnover and productivity distributions.

Paper 3

Campbell & Fisher (2000)

Aggregate Employment Fluctuations with Microeconomic Asymmetry

Authors: Jeffrey R. Campbell & Jonas D.M. Fisher

Journal: *American Economic Review* 90(5): 1323–1345

Year: 2000 **Field:** Macroeconomics; Labour economics

Classification: Structural (calibrated dynamic model + simulation)

3.1 Research Question

Can proportional plant-level costs of creating and destroying jobs—net adjustment costs—explain the empirically documented fact that the job destruction rate fluctuates more than the job creation rate in U.S. manufacturing?

3.2 Motivation

Davis et al. (1996) and Davis–Haltiwanger (1992) document that job destruction is substantially more volatile over the business cycle than job creation in U.S. manufacturing. This asymmetry challenges standard macroeconomic theory: the one-sector stochastic growth model with homogeneous plants predicts that job creation and destruction respond equally and symmetrically to aggregate shocks. Existing explanations either introduce arbitrary asymmetries in the driving process (Caballero 1992) or are quantitatively insufficient (Foote 1998). Campbell and Fisher ask whether a minimal, internally motivated modification—proportional *net* adjustment costs—can generate the observed asymmetry using a symmetrically distributed aggregate shock.

3.3 Main Contribution

The paper establishes that proportional net adjustment costs create an endogenous **policy asymmetry**: the employment decisions of job-destroying (shrinking) plants are more sensitive to wage changes than those of job-creating (expanding) plants. In a calibrated model with continuous idiosyncratic productivity shocks and ongoing aggregate wage fluctuations, this microeconomic asymmetry is preserved in the aggregate under plausible parameter values, generating job destruction volatility that exceeds job creation volatility—consistent with U.S. manufacturing data for the transportation equipment industry (SIC 37). The contribution is both analytical (establishing the mechanism) and quantitative (showing the aggregation condition under which it is preserved).

3.4 Relationship to the Literature

The paper sits at the intersection of the gross job flows literature (Davis–Haltiwanger 1990, 1992; Mortensen–Pissarides 1994; Caballero–Hammour 1994, 1996) and the employment adjustment cost literature (Sargent 1978; Hamermesh 1989; Bentolila–Bertola 1990; Hopenhayn–Rogerson 1993). Its distinctive feature is the emphasis on *net* adjustment costs: costs that depend on the *number* of jobs created or destroyed, not on the identity of the workers involved. Campbell and Fisher show that proportional (rather than quadratic or kinked) adjustment costs generate an asymmetry even with a symmetric driving process—a result unavailable from Caballero’s (1992) (S, s) framework.

3.5 Theoretical Framework and Economic Mechanism

Plant’s Problem

Each plant i produces output $z_t n_t^\alpha$ where z_t is idiosyncratic productivity (log-random-walk with i.i.d. innovation $\varepsilon_t \sim N(\mu, \sigma_z^2)$) and n_t is employment. The plant pays net adjustment costs: t_c units of lost output per job added and t_d units per job subtracted. Defining scaled variables $x = n_{t-1}/z^{1/(1-\alpha)}$ and $y = n_t/z^{1/(1-\alpha)}$, the Bellman equation is:

$$v(x, W) = \max_y \{y^\alpha - W y - \tau(y, x)(y - x) + \beta \mathbb{E}[u' v(y/u', W') \mid W]\}, \quad (3.1)$$

where $\tau(y, x) = t_c \mathbf{1}\{y > x\} - t_d \mathbf{1}\{y < x\}$ and $u' = (z'/z)^{1/(1-\alpha)}$.

Optimal Employment Policy

The solution defines two scaled thresholds, $y^-(W) < y^+(W)$:

$$\text{Create jobs: } n^* = y^-(W) z^{1/(1-\alpha)} \quad \text{if } n_{t-1} < n^-(z, W) \quad (3.2)$$

$$\text{Inaction: } n^* = n_{t-1} \quad \text{if } n^-(z, W) \leq n_{t-1} \leq n^+(z, W) \quad (3.3)$$

$$\text{Destroy jobs: } n^* = y^+(W) z^{1/(1-\alpha)} \quad \text{if } n_{t-1} > n^+(z, W) \quad (3.4)$$

Policy Asymmetry

The elasticities of the scaled employment thresholds with respect to W are:

$$\frac{\partial \ln y^-(W)}{\partial \ln W} = \frac{-1}{1-\alpha} \cdot \frac{W}{W + [1 - \beta(1-p)]t_c + \beta p t_d}, \quad (3.5)$$

$$\frac{\partial \ln y^+(W)}{\partial \ln W} = \frac{-1}{1-\alpha} \cdot \frac{W}{W - [1 - \beta(1-p)]t_d - \beta p t_c}, \quad (3.6)$$

where p is the probability of idiosyncratic-shock reversal. Because the denominator in (3.6) *subtracts* the adjustment cost terms while (3.5)’s denominator *adds* them, $|\partial \ln y^+ / \partial \ln W| > |\partial \ln y^- / \partial \ln W|$ for any strictly positive adjustment costs, regardless of their relative magnitudes.

Intuition: The total cost of the marginal job for a creating plant is $W + (\text{creation cost}) + \beta \cdot (\text{expected future destruction cost})$, which is *less* elastic to W than W alone. The total cost to a destroying plant of retaining the marginal job is $W - (\text{avoided destruction cost}) - \beta \cdot (\text{avoided future creation cost})$, which is *more* elastic to W . Hence, shrinking plants respond more strongly to any given wage change.

Aggregation and the Growth-Rate Asymmetry

The DH gross job flow formulas are:

$$\text{POS}_t = \frac{2N_{t-1}}{N_t + N_{t-1}} \int_0^1 \frac{n_{t-1}(i)}{N_{t-1}} g_t(i)^+ di, \quad (3.7)$$

$$\text{NEG}_t = \frac{2N_{t-1}}{N_t + N_{t-1}} \int_0^1 \frac{n_{t-1}(i)}{N_{t-1}} |g_t(i)^-| di. \quad (3.8)$$

A **growth-rate asymmetry** works *against* the policy asymmetry: because $\exp(\cdot)$ is convex, plants near the creation margin respond more strongly in actual growth rate terms than plants near the destruction margin, even when policy is symmetric in log terms. The net outcome depends on which force dominates—and policy asymmetry dominates when idiosyncratic shocks are *less* persistent.

3.6 Data

- **Calibration target:** Quarterly gross job creation and destruction data for SIC 37 (Transportation Equipment) from the LRD/BED.
- **Key moments:** Average net employment growth (-0.11% /quarter), average job reallocation rate (11.3%), standard deviation of employment growth (3.55%), first-order autocorrelation of employment growth (0.08).
- **Calibrated parameters:** $\alpha = 0.66$, $\beta = 1.052^{1/4}$ (5% annual discount); σ_z matched to job reallocation rate; aggregate shock parameters (ρ_W, σ_W) matched to employment growth moments.

3.7 Empirical Strategy

Calibration and model simulation, not estimation. The strategy:

1. Parameterise the model from standard values and calibrated moments for SIC 37.
2. Simulate the model to compute time series of aggregate POS and NEG.
3. Compare simulated moments (means, standard deviations, covariances of POS, NEG, NET) to their empirical counterparts.
4. Vary adjustment cost parameters (t_c, t_d) within plausible empirical bounds to assess sensitivity.

3.8 Identification Strategy

No formal identification in the econometric sense. Predictions are compared to data descriptively. The identifying variation is the parameterisation of adjustment

costs, taken from external evidence: Los Angeles employer surveys, Dutch firm studies (Hamermesh et al. 1994), Spanish structural estimates (Aguirregabiria–Alonso-Borrego 1998), and anecdotal evidence from aerospace production (Benkard 1998). The paper explicitly acknowledges that net adjustment costs are intrinsically difficult to measure, and that firm surveys typically measure *gross* adjustment costs (dependent on worker identity) rather than the *net* adjustment costs (dependent on number of positions) central to this model.

3.9 Main Results

Analytical Results (Section I)

- Policy asymmetry holds for *any* strictly positive adjustment costs, regardless of the relative magnitudes of t_c and t_d .
- The asymmetry is stronger when idiosyncratic shocks are *less* persistent (smaller ρ_{idio}), because the policy asymmetry dominates the growth-rate asymmetry.
- The growth-rate asymmetry always works against the policy asymmetry in computing aggregate flows.

Quantitative Results (Section III)

Model specification	std(NEG)/std(POS)	Match?
No adjustment costs ($t_c = t_d = 0$)	≈ 1.0	No
Baseline ($t_c = t_d = 0.5$, $\rho_W = 0.08$)	≈ 1.8 – 2.0	Yes
High persistence ($\rho_W = 0.5$)	< 1.0 in some specs	No
Empirical SIC 37	≈ 1.5 – 2.0	—

The model with baseline calibration generates a std(NEG)/std(POS) ratio comparable to the data. The cross-correlation $\rho(\text{POS}, \text{NEG})$ is negative and $\rho(\text{NET}, \text{SUM})$ is near zero, broadly consistent with data.

3.10 Economic Magnitude and Interpretation

The SIC 37 industry exhibits $\text{std}(\text{NEG}) \approx 2.0$ – 2.5% , $\text{std}(\text{POS}) \approx 1.0$ – 1.5% —a ratio of approximately 1.5–2.0. The model with plausible adjustment costs (roughly 0.5–1.5 quarterly wages) replicates this ratio. The mechanism operates through *aggregate* wage fluctuations, not idiosyncratic shocks—which by themselves generate symmetric flows. This is an important distinction: the paper identifies the aggregate component of gross job flow asymmetries.

3.11 Mechanisms

Three forces interact:

1. **Policy asymmetry (main driver):** Adjustment costs make the destruction threshold more elastic to wage changes than the creation threshold. Each 1%

increase in the wage causes shrinking plants to shrink by more than expanding plants expand.

2. **Growth-rate asymmetry (countervailing):** Convexity of actual vs. logarithmic growth rates makes job creators at the margin more responsive in actual terms, working against the policy asymmetry.
3. **Plant distribution dynamics:** Aggregation over the cross-section reflects the endogenous evolution of the distribution of scaled employment x across plants. With random-walk idiosyncratic shocks and proportional adjustment costs, this evolution preserves (rather than offsets) the policy asymmetry.

3.12 Robustness Checks

- Results reported for six different combinations of adjustment cost parameters and aggregate shock persistence.
- Model compared to economy-wide gross job flow data (not just SIC 37) with qualitatively similar results.
- General equilibrium version of the model developed (Section II.D) to show that the industry-equilibrium results have a valid GE interpretation, with the product wage driven by both labour supply and demand shocks.
- The paper explicitly notes that *gross* adjustment costs (costs that depend on worker identity, not net employment changes) would *not* generate the policy asymmetry, providing a discriminating prediction of the net vs. gross cost distinction.

3.13 Limitations

1. **No matching frictions:** Abstracts entirely from search and matching; the Mortensen–Pissarides mechanism (asymmetric matching vs. separation thresholds) could compound or offset the policy asymmetry.
2. **Symmetric aggregate shocks:** Assumes symmetrically distributed aggregate uncertainty; in reality, recessions may be more severe than expansions, compounding the asymmetry.
3. **Calibration, not estimation:** Adjustment costs are selected from a plausible range, not formally estimated. The model cannot be rejected on these dimensions.
4. **Manufacturing only:** Service sectors, where labour market dynamics may differ substantially, are excluded.
5. **No entry/exit:** The population of plants is fixed, abstracting from births and deaths that account for 20–25% of gross job flows in DH1992.

3.14 Policy Implications

- Firing costs (higher t_d) *reduce* employment volatility by dampening the destruction response, consistent with the Bentolila–Bertola (1990) motivation for European firing-cost policies—but at the cost of reduced flexibility.

- Policies reducing hiring costs specifically for expanding plants would raise the creation elasticity and reduce the volatility asymmetry.
- The paper provides a new rationale for the excess volatility of unemployment in recessions: it partly reflects the greater sensitivity of plant-level employment cuts to aggregate shocks, not just “cleansing” of less productive matches.

3.15 Overall Assessment

Campbell and Fisher (2000) is a rigorous paper that makes a genuinely original theoretical point—the policy asymmetry mechanism—and shows it has quantitative bite. The general equilibrium embedding is careful. The main weakness is the calibration approach, which limits the paper’s ability to formally assess how much of the observed asymmetry is explained by the mechanism. The paper is less influential than DH1992 partly because the Mortensen–Pissarides search-and-matching framework became dominant in the subsequent decade, and partly because the mechanism requires aggregate shocks to operate primarily through wages.

Quality: Good. A solid theoretical contribution with careful quantitative assessment.

3.16 Relevance to My Research

The policy asymmetry mechanism has direct relevance for understanding how Canadian firms respond to aggregate shocks (commodity price cycles, exchange rate changes) in different ways depending on whether they are expanding or contracting. The key testable prediction: the elasticity of employment cuts to adverse shocks should exceed the elasticity of employment additions to favourable shocks of the same magnitude, and this asymmetry should be larger for less persistent aggregate shocks (e.g., commodity price swings).

3.17 Potential Research Extensions

- Estimate t_c and t_d structurally from Canadian establishment-level data by matching moments of the job creation/destruction distribution, allowing estimation of the model’s degree of asymmetry.
- Test the model’s prediction that industries with more transient aggregate shocks (e.g., commodity sectors) show larger $\text{std}(\text{NEG})/\text{std}(\text{POS})$ ratios than industries with persistent shocks.

3.18 Research Gaps

- The paper does not model worker heterogeneity or worker-level welfare consequences of the asymmetric adjustment process.
- There is no explanation for why t_c and t_d should differ, or how they vary across industries, countries, or regulatory regimes.

3.19 Methodological Lessons

1. The DH growth rate metric versus the log-difference matters for quantitative model evaluation: the growth-rate asymmetry is a real force that works against the policy asymmetry and would be missed using log-approximations.
2. The policy asymmetry result holds *analytically* for any positive adjustment costs—showing a qualitative result before turning to quantitative assessment is methodologically important.
3. Comparing model-generated statistics to industry-level data (SIC 37) rather than economy-wide data is the right level of aggregation when the model is an industry-equilibrium model.

Five-Sentence Summary

Campbell and Fisher (2000) show that proportional net adjustment costs create a policy asymmetry at the plant level: shrinking plants are more sensitive to wage changes than expanding plants, because adjustment costs add to the total cost of job creation but reduce the total cost of job retention. This microeconomic asymmetry, when preserved through aggregation, generates a model in which the rate of job destruction fluctuates more than the rate of job creation in response to aggregate shocks—a key stylised fact from U.S. manufacturing that standard representative-agent models fail to reproduce. A growth-rate asymmetry (convexity of actual vs. log growth rates) works in the opposite direction and can dominate when idiosyncratic shocks are highly persistent, so the mechanism is strongest for relatively transient aggregate fluctuations. A calibrated version of the model for the U.S. transportation equipment industry generates a ratio $\text{std}(\text{NEG})/\text{std}(\text{POS})$ comparable to the data, explaining most of the observed excess volatility of job destruction without requiring an asymmetrically distributed driving process. The paper provides a clean, internally motivated explanation for job flow asymmetries based on the loss of production incurred when reorganising a plant to operate at a larger or smaller scale.

Quick Reference

Key contribution	Analytical policy asymmetry mechanism: net adjustment costs make job destruction more elastic to aggregate shocks than job creation, even with a symmetric driving process.
Key weakness	Calibration (not estimation) of adjustment costs; no entry/exit; model cannot be formally tested against alternative explanations.
Most interesting gap	Does the policy asymmetry vary systematically across industries with different regulatory environments (high vs. low firing costs)? A cross-country test of the mechanism would be revealing.
Publishable extension	Test the Campbell–Fisher mechanism using matched employer–employee data from Canada, comparing employment creation and destruction elasticities to province-level commodity price cycles across industries with high and low regulatory adjustment cost environments.

Paper 4

Davis & Haltiwanger (1999)

On the Driving Forces Behind Cyclical Movements in Employment and

Authors: Steven J. Davis & John Haltiwanger

Journal: *American Economic Review* 89(5): 1234–1258

Year: 1999 **Field:** Macroeconomics; Labour economics

Classification: Empirical (structural VAR)

4.1 Research Question

What types of structural disturbances—allocative shocks (which shift the distribution of employment opportunities across job sites) versus aggregate shocks (which move all plants in the same direction)—are the primary drivers of cyclical movements in aggregate employment and job reallocation in U.S. manufacturing?

4.2 Motivation

The DH (1992) paper documented that job reallocation is countercyclical and job destruction substantially more volatile than job creation. A key open question is whether these patterns reflect primarily allocative disturbances (sector-specific technology, oil price, or structural change shocks) or aggregate disturbances (monetary policy, aggregate demand, aggregate productivity). This distinction matters enormously for policy: if reallocation is primarily driven by allocative shocks, standard aggregate demand stabilisation cannot reduce it and might not even be desirable; if aggregate shocks drive most employment fluctuations, conventional macroeconomic interventions are more directly relevant. Previous VAR identification strategies (Blanchard–Quah 1989; Bernanke 1986) could not exploit the information in gross job flows. Davis–Haltiwanger (1999) develops a novel identification strategy using the creation–destruction decomposition that allows both qualitative and quantitative restrictions motivated by economic theory.

4.3 Main Contribution

1. **Identification via creation–destruction:** Develops a structural VAR where the decomposition of employment changes into creation and destruction rates provides novel identifying information unavailable in aggregate employment data alone. Aggregate and allocative disturbances have *qualitatively different* implications for creation and destruction: aggregate shocks move them in *opposite* directions, while allocative shocks move them in the *same* direction.

2. **Tighter inequality restrictions:** Theory implies that job destruction responds *at least as much* as job creation to aggregate shocks ($b_{na} \leq -1$), and that the contemporaneous impact of allocative shocks on job creation does not exceed that on job destruction ($|b_{ps}| \leq 1$). These restrictions, drawn from search-theoretic models and option-value arguments, substantially narrow the range of permissible inferences.
3. **Long historical series:** Estimates the system over 1947:Q1–1993:Q4, exploiting a long quarterly series constructed by splicing LRD data (1972–1993) with BLS turnover data (1947–1981) using a cyclically varying quit replacement rate.

4.4 Relationship to the Literature

The paper is a direct sequel to DH1992 and DH1990. It is methodologically related to the structural VAR literature (Blanchard–Quah 1989; King–Watson 1992; Shapiro–Watson 1988) but extends it by using the creation–destruction decomposition to generate additional identifying restrictions not available in aggregate time series. It contributes to the debate between “cleansing recession” theories (Caballero–Hammour 1994; Lilien 1982) and “sullyng” theories, and shows that the identification of which shock dominates depends critically on which restrictions one is willing to impose.

4.5 Theoretical Framework and Economic Mechanism

Structural Moving Average Representation

Let $Y_t = [\text{POS}_t, \text{NEG}_t]'$ and $\varepsilon_t = [\varepsilon_t^a, \varepsilon_t^s]'$ (aggregate and allocative structural innovations). The structural representation is:

$$Y_t = B(L) \varepsilon_t, \quad B(0) = B_0. \quad (4.1)$$

The estimated reduced-form VAR yields:

$$Y_t = D(L) \eta_t, \quad D(0) = I, \quad (4.2)$$

with reduced-form innovations $\eta_t = B_0 \varepsilon_t$.

Identification Problem

With diagonal elements of B_0 normalised to unity, there are two free parameters: b_{na} (contemporaneous effect of an aggregate shock on POS relative to NEG) and b_{ps} (contemporaneous effect of an allocative shock on NEG relative to POS). The zero-covariance restriction $\text{Cov}(\varepsilon^a, \varepsilon^s) = 0$ yields the moment equation:

$$b_{ps} = \frac{\sigma_{pn} - b_{na} \sigma_p^2}{\sigma_n^2 - b_{na} \sigma_{pn}}, \quad (4.3)$$

leaving the system one-dimensionally underidentified.

Weak Inequality Restrictions

- (i) $b_{na} < 0$: aggregate shocks move POS and NEG in opposite directions (4.4)
- (ii) $b_{ps} > 0$: allocative shocks move both in the same direction (4.5)

Tighter Theory-Motivated Restrictions

- (i)' $b_{na} \leq -1$: NEG responds at least as much as POS to aggregate shocks (4.6)
- (ii.a)' $|b_{ps}| \leq 1$: allocative shock impact on POS $\leq$ impact on NEG (4.7)
- (ii.b)' $\sum_{l=1}^m B_{11}(l) > 0$ for all $m \leq M=16$ (4.8)

Restriction (i)' follows from search models: separations can occur instantaneously while new matches require search time, so an aggregate shock causes a sharper contemporaneous rise in destruction than fall in creation. Restriction (ii.a)' follows from option value arguments: sunk costs of investment create an incentive to delay job creation in the face of uncertain future allocative disturbances, depressing creation even as destruction rises.

Long-Run Neutrality Restrictions (Alternative)

- (iv) $\sum_{l=0}^{\infty} [B_{11}(l) + B_{21}(l)] = 0$: aggregate shocks have no long-run effect on job reallocation (4.9)
- (v) $\sum_{l=0}^{\infty} [B_{12}(l) - B_{22}(l)] = 0$: allocative shocks have no long-run effect on employment (4.10)

4.6 Data

- **Primary (1972–1993):** LRD quarterly job creation and destruction data, updated through 1993 (Davis et al. 1996).
- **Secondary (1947–1981):** BLS monthly turnover data (accessions, layoffs, quits), converted using the Blanchard–Diamond (1990) method with a *cyclically varying* quit replacement rate.
- **Splicing:** Join the two series over their 1972–1981 overlap. ρ (implied NET growth, BLS 790 data) = 0.91.
- **Sample:** 1947:Q1–1993:Q4 (187 observations).

Statistic	POS	NEG	NET	SUM
Mean	5.8%	6.0%	−0.1%	11.8%
Std dev	1.2%	1.5%	2.0%	1.7%
Min	3.8%	2.9%	−5.9%	7.4%
Max	10.2%	10.8%	6.8%	16.3%

Source: BLS–LRD spliced series, 1947:Q1–1993:Q4

4.7 Empirical Strategy

Structural VAR estimation with multiple identification schemes:

1. Estimate a 2-variable VAR in POS and NEG with 4 lags (Dickey–Fuller tests reject unit roots in both series).
2. Derive the reduced-form covariance matrix.
3. Apply each set of identifying restrictions to recover admissible ranges for (b_{na}, b_{ps}) and corresponding variance decompositions.
4. Bootstrap standard errors using 1,000 Monte Carlo draws.
5. Construct historical decompositions at boundary values of the admissible parameter space (Figures 4–5).

4.8 Identification Strategy

This is the paper’s central methodological innovation. Rather than point- identifying the system, the paper derives *ranges* of parameter values consistent with a set of qualitative restrictions, then reports the implied range of variance decompositions.

Key identification elements:

- **Sign restrictions on contemporaneous responses:** (i) and (ii) are definitional and accepted by nearly all theories.
- **Magnitude restrictions:** (i)’ and (ii.a)’ are theory-motivated from search models and option value arguments; these are the *binding* restrictions that narrow the admissible range.
- **Zero covariance:** Standard in the VAR literature; the paper explores sensitivity to relaxing this (Figures 7–8).

Critical limitation: The mapping from (b_{na}, b_{ps}) to variance decompositions is highly nonlinear (Figure 3). The allocation of variance between aggregate and allocative shocks is extremely sensitive to where in the admissible range one sits. The paper honestly reports ranges rather than point estimates—an intellectually rigorous but informationally limited strategy.

4.9 Main Results

Under Weak Restrictions (Figure 3)

Admissible range: $b_{na} \in (-2.5, -0.87)$.

- Allocative shocks account for 5–20% of 4-step forecast-error variance of employment growth.
- Aggregate shocks are always a major driving force behind employment fluctuations.
- Allocative shocks account for 20–90% of job reallocation variance (wide range).

Under Tighter Restrictions (Table 3, Columns 1–2)

Admissible range narrows to $b_{na} \in (-1.63, -1.0)$, $b_{ps} \in (0.1, 1.0)$.

Variable	Range (4-step horizon)	Range (16-step horizon)
Employment growth	3%–13%	5%–17%
Job reallocation	44%–80%	49%–84%

Fraction of forecast-error variance explained by *allocative* shocks.

Key inference: Allocative disturbances are a *minor* driver of employment fluctuations but the *dominant* driver of job reallocation intensity.

Under Neutrality Restrictions (Table 3, Columns 3–5)

- Restricting allocative shocks to have no long-run employment effect (v) yields $b_{na} = 0.03$, $b_{ps} = -0.41$ —*violates* weak qualitative restrictions (i), (ii), (i)'.
- Restricting aggregate shocks to have no long-run reallocation effect (iv) yields $b_{na} = -0.68$, $b_{ps} = -0.12$ —violates (i)' and (i).
- Under these neutrality restrictions, allocative shocks account for 34–80% of employment variance and >80–94% of reallocation variance.

4.10 Economic Magnitude and Interpretation

The paper's core quantitative finding under the tighter qualitative restrictions is:

- **Aggregate shocks account for 83–97%** of cyclical variation in employment growth (4-step forecast errors).
- **Allocative shocks account for 44–80%** of cyclical variation in job reallocation intensity.

These two findings are not in tension: aggregate shocks cause recessions; allocative shocks cause the gross flows that accompany them independently. The economy's output cycles and its reallocation cycles are driven by different forces.

4.11 Mechanisms

Two mechanisms determine the relative importance of the two shock types:

1. **Opportunity cost mechanism** (Davis–Haltiwanger 1990): during recessions, the opportunity cost of relocating workers falls, so the pace of allocative reallocation endogenously increases. A constant stream of allocative disturbances generates countercyclical reallocation intensity.
2. **Option value / sunk cost mechanism:** The sunk nature of investments required to create new vacancies generates an option value for waiting. When allocative shocks are serially correlated (Table 2 shows significant autocorrelations in cross-industry return dispersion at lags 1–6), this option value simultaneously depresses job creation and boosts job destruction—enabling $b_{ps} < 0$, i.e., allocative innovations that cause short-run employment declines.

4.12 Robustness Checks

- Results reported for five different identification schemes (qualitative and neutrality restrictions).
- Sensitivity analysis for $\rho(\varepsilon^a, \varepsilon^s) \in \{-0.3, 0, +0.3\}$ (Figures 7–8).
- BLS 790 data cross-validation ($\rho = 0.91$ between implied and published net employment growth).
- Dickey–Fuller tests confirm stationarity of POS and NEG series.
- VARs with deterministic trends as sensitivity checks.

4.13 Limitations

1. **One-dimensional underidentification:** No single restriction produces a unique parameter point; results are necessarily range-valued.
2. **Two-variable system:** Cannot distinguish among multiple types of allocative or aggregate shocks (technology, monetary, fiscal, oil).
3. **Manufacturing only:** The long historical series is available only for manufacturing, which exhibits secular decline over the sample.
4. **BLS splicing:** The 1947–1971 portion relies on a maintained assumption about quit replacement rates, introducing potential measurement error.
5. **Linearity:** Nonlinear interactions between aggregate and allocative shocks—which the theories all suggest—are not accommodated.
6. **Inconsistency of neutrality and qualitative restrictions:** Long-run neutrality restrictions produce (b_{na}, b_{ps}) values that violate the sign restrictions, revealing internal inconsistency and limiting the credibility of the long-run results.

4.14 Policy Implications

- Because aggregate shocks drive most cyclical employment fluctuations (under preferred tighter restrictions), standard aggregate demand policies can influence employment—but do not address the underlying allocative process that determines job reallocation intensity.

- The finding that allocative shocks are a minor driver of employment cycles but a dominant driver of reallocation suggests that policies targeting “structural unemployment” address a phenomenon largely independent of macroeconomic stabilisation.
- The long-run neutrality results—where allocative shocks account for 34–80% of employment variance—would, if taken seriously, suggest a much larger role for structural policies. However, these results are theoretically inconsistent with the qualitative restrictions.

4.15 Overall Assessment

Davis–Haltiwanger (1999) is a methodologically important paper that introduces a genuinely novel approach to structural VAR identification using the creation–destruction decomposition. It is honest about the limits of what the data can identify and clearly maps the trade-offs between different sets of assumptions. The main empirical finding—under theory-motivated tighter restrictions, aggregate shocks drive most employment cyclicalities while allocative shocks drive most reallocation cyclicalities—is sensible and has influenced the subsequent literature. The paper’s weakness is that its conclusions are range-valued rather than point estimates, and the ranges are wide enough to be somewhat uninformative on policy-relevant questions.

Quality: Very good. A methodologically careful paper with important empirical findings, though limited by data and identification constraints.

4.16 Relevance to My Research

The identification strategy—using the creation–destruction decomposition to distinguish allocative from aggregate shocks—is directly applicable to Canadian data. With LEAP/LEED, one could construct quarterly job creation and destruction series for the entire Canadian economy and estimate the DH1999 structural VAR. The Canadian context is particularly interesting because the economy faces both aggregate shocks (U.S. monetary policy spillovers, global demand) and strong allocative shocks (commodity price cycles with large sector-specific effects). Testing whether the relative importance of allocative vs. aggregate shocks differs in resource-intensive provinces versus manufacturing provinces would be a natural extension.

4.17 Potential Research Extensions

- Extend the DH1999 VAR to the service sector in Canada using the full LEAP/LEED universe, which would make the sample more representative of the aggregate economy.
- Add commodity price innovations and exchange rate movements as additional variables in an expanded VAR, exploiting Canadian resource-sector dynamics to better identify allocative shock contributions.
- Use regional variation in commodity price exposure to construct province-specific identifying restrictions on allocative shock magnitudes.

4.18 Research Gaps

- The paper uses only two variables (POS, NEG). Expanding to a four-variable system (POS, NEG, unemployment rate, vacancy rate) would permit Beveridge curve restrictions and substantially tighten identification.
- The allocation between aggregate and allocative shocks is sensitive to $\rho(\varepsilon^a, \varepsilon^s)$ but the paper has no empirical estimate of this parameter.

4.19 Methodological Lessons

1. The creation–destruction decomposition provides identifying information unavailable in aggregate net employment data: allocative and aggregate shocks have qualitatively different creation–destruction responses, and theory imposes magnitude restrictions.
2. When a system is underidentified, reporting the range of inferences over the admissible parameter space (rather than arbitrary point identification) is the intellectually honest approach.
3. Long-run neutrality restrictions and short-run qualitative restrictions can be internally inconsistent (Table 3), providing a diagnostic for whether theoretical identifying assumptions are self-consistent.
4. The serial correlation of cross-industry stock return dispersion (Table 2) as indirect evidence for the option value mechanism is a simple and elegant test of a theoretical underpinning of an identifying assumption.

Five-Sentence Summary

Davis and Haltiwanger (1999) develop a structural VAR framework for a bivariate system of quarterly job creation and destruction rates in U.S. manufacturing (1947–1993), exploiting the qualitatively different responses of creation and destruction to aggregate versus allocative shocks to generate novel identifying restrictions unavailable in aggregate employment data alone. Weak inequality restrictions establish that aggregate shocks are always major driving forces behind employment fluctuations, while the contribution of allocative shocks to employment variance ranges widely (5–20%) depending on the structural parameter values. Imposing tighter theory-motivated restrictions—that job destruction responds at least as much as creation to aggregate shocks, and that allocative shocks cannot simultaneously boost employment—narrows the range to 3–17% of employment variance explained by allocative shocks but confirms that allocative shocks account for 44–80% of job reallocation intensity variance. Long-run neutrality restrictions imply an even larger role for allocative shocks in both employment and reallocation variance, but these restrictions are internally inconsistent with the qualitative sign restrictions. The paper’s core message is that aggregate shocks drive most cyclical employment fluctuations, while allocative disturbances are the dominant driver of the countercyclical movements in job reallocation that accompany those cycles.

Quick Reference

Key contribution	Novel structural VAR identification using job creation–destruction decomposition; maps the full range of inferences over the admissible structural parameter space under qualitative restrictions from theory.
Key weakness	System remains one-dimensionally underidentified; wide admissible ranges limit sharp empirical conclusions; manufacturing-only sample; long-run and short-run restrictions are internally inconsistent.
Most interesting gap	Expanding to a four-variable system (POS, NEG, unemployment, vacancies) with Beveridge curve restrictions would tighten identification further and address whether the allocative–aggregate decomposition is stable across business cycle phases.
Publishable extension	Construct a DH1999-type structural VAR for the full Canadian economy using quarterly LEAP/LEED-based creation and destruction series, with an expanded variable set including commodity price indices to separately identify resource-sector allocative shocks.

Paper 5

Lazear & Spletzer (2012)

Hiring, Churn, and the Business Cycle

Authors: Edward P. Lazear & James R. Spletzer

Journal: *American Economic Review: Papers & Proceedings* 102(3): 575–579

Year: 2012 **Field:** Labour economics; Macroeconomics

Classification: Empirical (descriptive accounting decomposition)

5.1 Research Question

How does employment churn—the simultaneous hiring and separation within a business that offsets each other without changing net employment—vary over the business cycle, and what role did changes in churn play in the collapse of hiring during the 2007–09 Great Recession?

5.2 Motivation

Standard analyses of the Great Recession focused on net employment (the dramatic fall in payrolls) and on job creation and destruction. But these net measures obscure a potentially important channel: the cessation of *churn*. Even businesses that are not changing their employment level engage in substantial ongoing hiring and separation activity—workers quit, are hired as replacements, transfer, or retire. This churn is intrinsically valuable: it enables workers to move to more productive jobs and enables firms to replace mismatched workers. If churn collapses during recessions, labour market efficiency losses may substantially exceed what net employment changes alone reveal. Lazear and Spletzer construct the first explicit churn decomposition from JOLTS microdata to quantify this channel.

5.3 Main Contribution

The paper introduces an explicit accounting decomposition of total hires and separations into four components—growth hires (H_E^G), replacement hires (churn in expanding firms), replacement separations (churn in contracting firms), and employment-decreasing separations—and uses JOLTS microdata to estimate each component quarterly. Key findings: churn is (i) procyclical ($\rho(\text{churn}, u) = -0.96$); (ii) quantitatively enormous (65% of all hires in normal times were churn); and (iii) the dominant component of the hiring collapse during the 2007–09 recession (**79%** of the decline in hires was due to reduced churn, not reduced job creation).

5.4 Relationship to the Literature

The paper fits into the wider gross flows literature initiated by DH1992 but focuses on within-firm turnover (churn) rather than across-firm flows (job creation/destruction). It is related to Anderson–Meyer (1994), Burgess–Lane–Stevens (2000), and Davis–Faberman–Haltiwanger (2006), who had estimated that 42–70% of hiring was churn using various data sources. The paper’s contribution is to use JOLTS *microdata*—enabling the within-establishment decomposition—and to examine the business-cycle variation directly. Churn is distinct from, but related to, the “excess reallocation” concept in DH1992.

5.5 Theoretical Framework and Economic Mechanism

Accounting Framework

For an establishment in period t , define hires H and separations S in business of type $j \in \{E = \text{expanding}, C = \text{contracting}, Z = \text{zero-growth}\}$. Total hires and separations decompose as:

$$H = H_E^G + CH, \quad (5.1)$$

$$S = S_C^D + CH, \quad (5.2)$$

where $CH = CH_E + CH_C + CH_Z$ is total churn:

$$CH_E = H_E^R = S_E \quad (\text{replacement hires in expanding firms}) \quad (5.3)$$

$$CH_C = H_C = S_C^R \quad (\text{replacement separations in contracting firms}) \quad (5.4)$$

$$CH_Z = H_Z = S_Z \quad (\text{all hires/seps in zero-growth firms}) \quad (5.5)$$

Net employment change follows directly:

$$H - S = H_E^G - S_C^D. \quad (5.6)$$

Crucially, total churn CH cancels exactly from the net change equation. Churn has no direct effect on net employment—but it determines the *level* of total hiring activity.

Two Cyclical Mechanisms

1. **Demand side:** In downturns, businesses decrease their hiring and separations that would have been matched by replacement hires remain unfilled $\Rightarrow$ churn falls as unfilled separations become employment-decreasing separations.
2. **Supply side:** Workers become reluctant to quit in recessions (fear of unemployment) $\Rightarrow$ quit rate falls $\Rightarrow$ fewer replacement hires needed $\Rightarrow$ churn falls through a different channel.

5.6 Data

- **Dataset:** JOLTS (Job Openings and Labour Turnover Survey) microdata, U.S. Bureau of Labour Statistics.

- **Coverage:** Random sample of $\approx 16,000$ nonfarm business establishments; $\approx 10,500$ provide data regularly.
- **Period:** December 2000 – June 2011; estimated at quarterly frequency.
- **Key feature:** JOLTS microdata allow classification of each establishment by its net employment change each quarter (expanding, contracting, zero-growth), enabling the within-establishment decomposition.
- **Limitation:** Excludes imputed hires and separations for unobserved births and deaths; quarterly estimates (≈ 12 m hires in mid-2000s) are below published monthly statistics (≈ 15 m) partly for this reason.

5.7 Empirical Strategy

Pure descriptive decomposition. The strategy:

1. Classify each JOLTS establishment in each quarter as expanding ($\Delta E > 0$), contracting ($\Delta E < 0$), or zero-growth ($\Delta E = 0$).
2. Within each category, compute hires and separations from JOLTS microdata.
3. Apply the churn accounting identities (5.1)–(5.6) to decompose total hires and separations.
4. Compute seasonally adjusted time series for each component (Figure 1).
5. Formally decompose the change in churn from 2007:Q4 to 2009:Q1 into: (a) change in the number of establishments with churn; (b) change in average size of churn conditional on having churn (Table 1).

5.8 Identification Strategy

No causal identification. This is a descriptive accounting exercise. The decomposition is an identity, not a regression. The paper makes no claim that churn *causes* employment changes or that the recession *caused* the decline in churn—it documents the accounting fact that 79% of the decline in hires during 2007–09 was attributable to a decline in churn. The welfare implications (cost ≈ 0.4 pp of GDP annually) rely on auxiliary assumptions about the productivity value of worker reallocation via churn, not on causal identification.

5.9 Main Results

Stylised Facts about Churn

Statistic	Value
Share of hires that is churn (mid-2000s)	65%
$\rho(\text{churn}, u)$	−0.96
$\rho(\text{churn}, \text{NET})$	+0.43 (2001–2011:Q2)
$\rho(\text{churn}, \text{NET})$	+0.79 (2001–2009:Q2)

Great Recession Decomposition

Component	2007:Q4	2009:Q2
Total churn (millions)	8.3	5.3
Total hires (millions)	≈ 12	≈ 8.2
<i>Decomposition of the 3.0m decline in churn</i>		
Fewer establishments with churn	1.9m $\rightarrow$ 1.6m	52%
Smaller average churn size	4.44 $\rightarrow$ 3.78 per estab.	48%
Share of hiring decline due to churn	79%	
Share due to lower job creation (H_E^G)	$\approx 25\%$	

Source: JOLTS microdata, seasonally adjusted.

5.10 Economic Magnitude and Interpretation

The finding that 79% of the Great Recession’s hiring collapse was churn-driven is the headline result and represents a major reframing of what happened in the 2007–09 recession. Standard analyses focused on job creation collapsing and job destruction rising. Lazear–Spletzer show that these forces, while real, were quantitatively swamped by the simultaneous collapse of the replacement-hire/quit mechanism. The economy became *frozen* rather than simply contracting in the traditional sense.

The welfare cost estimate:

- Reduced churn ≈ 0.4 percentage points of GDP *annually* over the 3.5 years following the recession onset.
- This estimate uses the auxiliary assumption that churn-driven worker reallocation raises productivity by observed average productivity differences between gaining and losing firms.

5.11 Mechanisms

Two mechanisms are empirically consistent but the paper cannot distinguish between them:

1. **Supply side (worker reluctance):** Workers become reluctant to quit in recessions (fear of being unable to find new employment), so the quit rate falls. This reduces replacement hiring even at firms that would gladly replace a quitter.
2. **Demand side (employer frugality):** Employers, when a worker separates, choose not to replace them—tolerating temporary staffing gaps. This mechanism is the paper’s preferred interpretation: “separations that would have been matched by hiring during good times remain unfilled.” The causal chain runs: aggregate shock $\rightarrow$ employer decision not to replace $\rightarrow$ reduced churn.

The 52/48 extensive/intensive margin split of the churn decline is consistent with both mechanisms.

5.12 Robustness Checks

- Validation of JOLTS microdata estimates against published JOLTS statistics (hires and separations track published series).
- Validation of job creation/destruction estimates against BLS Business Employment Dynamics (BED) data.
- Consistency of the 65% churn share with prior estimates (Anderson–Meyer 1994: 69%; Burgess–Lane–Stevens 2000: $\approx 70\%$; Davis et al. 2006: 42–72%).

Note that exclusion of births/deaths imputation is a consistent understatement and does not affect the proportional decomposition.

5.13 Limitations

1. **JOLTS sample:** $\approx 10,500$ establishments, disproportionately large businesses. Small businesses, which tend to have higher churn rates, may be underrepresented.
2. **Short time series:** JOLTS microdata begin in December 2000; only one full business cycle is available, with the 2001 recession only partially captured.
3. **No worker-level data:** Churn is measured at the establishment level. It is impossible to identify whether replacement hires are the same or different workers from those who separated—which matters for productivity analysis.
4. **Nonfarm private sector only:** Government employment is excluded; education and health services differ substantially in quit behaviour.
5. **No causal identification:** The paper documents the accounting fact that churn declined. Whether this represents an efficiency loss or a rational adjustment to changed incentives is not answered.
6. **P&P format:** As a 5-page Papers & Proceedings paper, many analytical details are necessarily compressed. This is a short report of findings rather than a full research paper.

5.14 Policy Implications

- If 79% of the Great Recession’s hiring collapse was churn-related, then policies designed to stimulate job creation (investment tax credits, hiring subsidies) would have addressed only $\approx 25\%$ of the hiring shortfall.
- Policies that encourage worker mobility (portable benefits, reductions in non-wage costs of switching jobs) would target the mechanism directly.
- The $\approx 0.4\%$ of GDP welfare cost from reduced churn provides a rough benchmark for evaluating policies that restore labour market fluidity.
- Aggregate demand stimulus might restore churn partly by restoring worker confidence to quit—a labour supply channel distinct from the job creation/destruction channel.

5.15 Overall Assessment

This is an insightful, well-executed descriptive paper that reframes a key fact about the Great Recession. The accounting decomposition is clean and the empirical finding is striking. Its limitations as a short P&P paper are acknowledged: it establishes a stylised fact rather than a causal mechanism, and the welfare calculation rests on auxiliary assumptions that are not formally derived. The paper raises more questions than it answers—the mechanisms behind the churn collapse, the welfare implications, the heterogeneity across industries and workers—which is entirely appropriate for a conference proceedings contribution that motivates a more complete follow-up.

Quality: Good for its format. Concise, clear, empirically novel, and policy-relevant.

5.16 Relevance to My Research

Very relevant for RDC work on Canadian labour dynamics. The JOLTS-based decomposition could be replicated using Canadian administrative data (LEED or the Longitudinal Worker File), which allows similar classification of establishments by their employment direction. The hypothesis that churn is procyclical and was severely disrupted during the 2008–09 recession and the COVID-19 pandemic in Canada could be tested directly. Of particular interest would be whether the sectoral composition of churn collapse differs in Canada: given the resource sector’s sensitivity to commodity cycles, churn dynamics in Alberta and Saskatchewan would likely differ sharply from Ontario and Québec during commodity busts.

5.17 Potential Research Extensions

- Replicate the Lazear–Spletzer decomposition for Canada using LWF/LEED data for 2000–2024, covering the 2008–09 recession and the COVID-19 shock.
- Disaggregate churn by industry, province, worker age, and occupation to identify which segments of the Canadian labour market experienced the largest efficiency losses from churn collapse.
- Estimate a structural model of the quit/hire decision to formally identify whether the churn collapse in Canada was supply-driven (workers reluctant to quit) or demand-driven (employers declining to replace quitters), using variation in provincial unemployment insurance generosity as an instrument for worker quit behaviour.

5.18 Research Gaps

- The paper uses establishment-level data but cannot identify whether churn involves high-to-low or low-to-high productivity transitions. A key question is whether *efficient* churn collapsed during the recession or whether the decline was concentrated in inefficient (mismatch-correcting) turnover.
- The cost estimate (0.4% of GDP/year) is rough and does not account for the possibility that some churn during booms was excessive (overhiring/overquitting driven by optimism), so not all lost churn represents an efficiency loss.

5.19 Methodological Lessons

1. A simple accounting decomposition can reveal economically important facts invisible in aggregate statistics—the 79% finding emerges directly from the churn identity applied to microdata and would be invisible in aggregate employment or JOLTS-level statistics.
2. Classifying establishments by employment *direction* (expanding/contracting/zero-growth) within each period, rather than by long-run growth status, is essential for implementing the churn decomposition correctly.
3. Validating microdata-based estimates against published aggregate statistics (BED, published JOLTS) provides reassurance that microdata classifications are consistent with external benchmarks.

Five-Sentence Summary

Lazear and Spletzer (2012) use JOLTS establishment microdata to decompose total hires and separations into four components: growth hires (job creation), employment-decreasing separations (job destruction), and churn (replacement hires and separations that offset each other within a firm without changing net employment). They show that churn accounts for approximately 65% of all hiring in normal times and is strongly procyclical, with a correlation of -0.96 with the unemployment rate over 2001–2011. During the 2007–09 Great Recession, churn fell by 3.0 million workers per quarter (36% decline), and this churn collapse accounts for 79% of the total decline in hiring—far larger than the decline in job creation. The decline in churn was evenly split between fewer establishments experiencing any churn (52%) and smaller average churn among those that did (48%), consistent with a combination of worker reluctance to quit and employer decisions to leave separations unfilled. The welfare cost of this reduced churn—lost productivity from foregone efficient worker reallocation—is estimated at roughly 0.4 percentage points of GDP annually over the post-recession period, highlighting that the efficiency costs of recessions substantially exceed those revealed by net employment changes alone.

Quick Reference

Key contribution	First explicit churn decomposition from JOLTS microdata; documents that 79% of the Great Recession's hiring collapse was churn-driven rather than job-creation-driven—fundamentally reframing the labour market contraction.
Key weakness	Short P&P format; no causal identification; cannot distinguish supply-side (reluctant quitters) from demand-side (non-replacing employers) mechanisms; no worker-level data to identify productivity effects of churn.
Most interesting gap	Whether the churn collapse was efficient (eliminating mismatch-driven excessive turnover) or inefficient (preventing productive reallocation)—the answer determines whether recession-period policy should try to restore churn or accept it as rational adjustment.
Publishable extension	A full-length paper using Canadian LWF/LEED data to replicate the churn decomposition over 2000–2024, disaggregating by province, industry, age, and occupation, and formally testing the supply vs. demand mechanism using provincial unemployment insurance generosity as an instrument for worker quit behaviour.

Paper 6

Hyatt & McEntarfer (2012)

Job-to-Job Flows in the Great Recession

Authors: Henry Hyatt & Erika McEntarfer (U.S. Census Bureau, CES)

Journal: *American Economic Review: Papers & Proceedings* 102(3): 580–583

Year: 2012 **Field:** Labour economics; Macroeconomics

Classification: Empirical (descriptive measurement)

6.1 Research Question

How did flows of workers between jobs—both direct job-to-job transitions and transitions involving nonemployment—change during the Great Recession, and what happened to the earnings changes associated with those flows?

6.2 Motivation

Despite extensive documentation of aggregate unemployment dynamics during the Great Recession, relatively little was known about the underlying worker- level transitions—specifically, how the *type* of job separation (direct job-to-job versus separation through nonemployment) changed in severity and the earnings consequences for each type. Standard labour-market surveys cannot trace workers across employers. Two prior strands of work—Jacobson, Lalonde, and Sullivan (1993) on earnings losses from displacement, and Topel and Ward (1992) on early-career wage growth from job change—had documented that job change matters for lifetime earnings but had not connected these to the cyclical behaviour of transition types. Hyatt and McEntarfer construct the first nationally-anchored longitudinal employer-employee measures of distinct job-to-job flow categories to characterise labour market adjustment during the Great Recession.

6.3 Main Contribution

1. **Direct job-to-job (J2J) flows are strongly procyclical** and fell to a 12-year series low by early 2009, preceding the official recession start by 1–2 quarters. Separations to long nonemployment spells are countercyclical.
2. **Earnings changes from job transitions are procyclical** at every transition type; in the Great Recession all transition types hit series lows and the earnings *penalty* for nonemployment widened.
3. **Housing bust case study:** Residential construction workers experienced a sharp J2J collapse (flows fell to 70% of their 2004–06 level), much higher rates

of industry switching (two-thirds leaving construction), and median earnings losses of -3.4% for inter-industry movers versus small gains in prior periods.

6.4 Relationship to the Literature

The paper sits at the intersection of (i) gross worker flows (Blanchard–Diamond 1990; Davis–Haltiwanger 1992), (ii) J2J flow measurement (Fallick–Fleishman 2004; Bjelland et al. 2011), and (iii) earnings dynamics and job change (Jacobson, Lalonde, Sullivan 1993; Topel–Ward 1992). The contribution is combining all three in a quarterly administrative data framework that separates transition types by nonemployment duration. It is the empirical predecessor to the Census Bureau’s full J2J statistics programme.

6.5 Theoretical Framework and Economic Mechanism

No formal model is presented. The organising framework rests on two forces:

- **Procyclical voluntary quits:** In good times, workers make frequent J2J moves to capture wage gains up a job ladder; in recessions workers are reluctant to quit, collapsing J2J flows.
- **Earnings penalty amplified by spell duration:** Longer nonemployment spells generate larger earnings losses at reemployment, consistent with human capital depreciation, stigma, or mismatch.

6.6 Data

- **Dataset:** LEHD (Longitudinal Employer-Household Dynamics) multi-state pilot database.
- **Frame:** 9 anchor states (CA, FL, GA, IL, KS, MI, NV, NC, ND); national job histories constructed for these workers covering flows to/from >40 states.
- **Period:** 1998–2010 (quarterly).
- **Flow types:** (1) J2J same quarter; (2) J2J adjacent quarter; (3) nonemployment 1 quarter; (4) nonemployment >1 quarter; (5) dominant separations (total).
- **Limitation:** Cannot distinguish unemployed from out of labour force; nonemployment durations only approximately measured in quarterly earnings data.

6.7 Empirical Strategy

Pure descriptive measurement:

1. Classify separators by the duration of the intervening nonemployment spell between primary jobs.
2. Compute seasonally adjusted quarterly time series for each flow type (Figure 1) and median earnings changes by transition type (Figure 2).
3. Case study of residential building construction (NAICS 2361): compute frequency, industry destination, and earnings changes for J2J flows originating in

construction across three periods (2001–03, 2004–06, 2007–09).

6.8 Identification Strategy

No causal identification. Pure descriptive measurement. The comparison across periods for residential construction (Table 1) is a descriptive before/after comparison, not an event study with a control group.

6.9 Main Results

Aggregate J2J Flows (Figure 1)

- Direct J2J flows (same and adjacent quarters) began falling in early 2007—1–2 quarters *before* the official recession start (NBER December 2007).
- Both types fell to a 12-year series low by early 2009.
- Separations to >1 quarter of nonemployment: slight countercyclical pattern; large spike in late 2008 driven almost entirely by spells lasting ≥ 1 year.

Earnings Changes from Job Transitions (Figure 2)

Transition type	2006:Q2 (boom)	Recession
Direct J2J (same quarter)	+9%	Series low; penalty for NE widened
J2J (adjacent quarter)	+3.8%	Procyclical decline
Nonemployment 1 quarter	0%	Negative
Nonemployment 2–3 quarters	–1.2%	Larger losses than 2001 recession

All transition types show procyclical comovement in earnings changes.

Residential Construction Case Study (Table 1)

Measure	2004–06	2007–09
J2J flows (index)	100	≈ 70
Share staying in construction	$\sim 41\%$	$\sim 37\%$ (two-thirds left)
Median wage change, within construction	+5.2%	–0.6%
Median wage change, cross-sector	+2.7%	–3.4%
Share flowing to Accommodation & Food	7.9%	10.0%

Accommodation & Food Services movements associated with 13–21% downward earnings moves.

6.10 Economic Magnitude and Interpretation

The collapse in direct J2J flows begins before the NBER recession, suggesting the labour market “felt” the downturn through job-switching channels first. The 30% fall in construction J2J flows implies that roughly one-third of the job-changing activity

that normally keeps construction workers on career ladders stopped entirely. The earnings penalties—up to -10.6% for retail trade movers; -20% for accommodation movers from construction—illustrate how sectoral reallocation during the housing bust imposed large permanent earnings losses.

6.11 Mechanisms

1. **Procyclical voluntary quits:** J2J collapse reflects workers ceasing to quit voluntarily in recessions, not a change in the gross separation rate.
2. **Earnings penalty amplified by nonemployment duration:** Consistent with human capital depreciation, stigma, or mismatch (workers forced into lower-quality matches under duress).

6.12 Robustness Checks

Short P&P paper; no explicit robustness section. The construction of J2J flow categories follows established conventions from Bjelland et al. (2011) and Fallick, Haltiwanger, and McEntarfer (2011). Restriction to *primary* (maximum earnings) jobs is motivated by prior evidence that most meaningful J2J flows occur between primary jobs.

6.13 Limitations

1. **P&P format (4 pages):** No regression analysis, worker characteristic controls, or formal inference.
2. **Nine-state frame:** Not nationally representative.
3. **Cannot distinguish U from N:** LEHD shows only earnings.
4. **Approximate nonemployment durations:** One full quarter of nonemployment = 3–8 months in reality.
5. **No causality:** Housing bust case study is descriptive.

6.14 Policy Implications

- J2J flows begin declining before official NBER recession start $\Rightarrow$ fluidity measures should complement unemployment rate as leading labour market indicators.
- Large earnings losses for construction workers (-3.4% cross-sector; -20% to accommodation) support targeted retraining policies for displaced construction workers.
- J2J flows are the primary vehicle for early-career wage growth (Topel–Ward 1992); their cyclical collapse implies recession-scarring effects on youth wage trajectories.

6.15 Overall Assessment

A high-quality descriptive paper that opens up an important new data source and establishes clean stylised facts about how the *type* of worker transition changes over

the business cycle. Its strength is the data construction and clarity of empirical patterns. Its weakness is the P&P format which prevents deeper analysis. The findings motivated the full Census Bureau J2J statistics programme.

Quality: Good for its format. Clean measurement, novel facts, policy-relevant.

6.16 Relevance to My Research

Directly relevant for Canadian RDC work. LEED or the Longitudinal Worker File can be used to construct analogous J2J flow series for Canada. Testing whether Canadian J2J flows are equally procyclical, and whether the Great Recession or commodity bust produced similar collapse dynamics in resource provinces, would be a natural paper.

6.17 Potential Research Extensions

- Construct Canadian J2J flow series from LEED 2000–2024; compare procyclicality magnitude to U.S. and across provinces.
- Study J2J flows from resource-sector workers during commodity busts (2014–16 oil price crash; COVID-19).

6.18 Research Gaps

- No formal distinction of supply-side (workers not quitting) vs. demand-side (employers not hiring) mechanisms.
- No analysis of how J2J flows differ by worker age, education, or tenure—all relevant for career-ladder interpretations.

6.19 Methodological Lessons

1. Restricting J2J flow measurement to *primary* jobs (highest quarterly earnings) is crucial to avoid conflating secondary jobs with meaningful career transitions.
2. Separating transitions by nonemployment duration (0, 1, >1 quarter) reveals very different cyclical patterns across types.
3. Leading indicator property: J2J flows decline 1–2 quarters before the official NBER recession date.

Five-Sentence Summary

Hyatt and McEntarfer (2012) use a nine-state LEHD pilot database to construct quarterly measures of four types of worker transitions—direct job-to-job, short nonemployment, and long nonemployment—for the period 1998–2010. They show that direct J2J flows are strongly procyclical, began declining in early 2007 before the official recession start, and fell to a 12-year low by early 2009, while separations to long nonemployment spells are countercyclical. Earnings changes associated with all job transitions are procyclical, with direct J2J movers earning $\approx 9\%$ more at reemployment in booms versus near zero in recessions; the penalty for nonemployment widened in 2007–09. A case study of residential construction reveals that J2J flows fell to $\approx 70\%$

of their pre-recession level, two-thirds of movers left the sector, and median earnings changes turned negative across all destinations. The paper establishes that direct J2J flows—the primary vehicle for career advancement—collapse disproportionately in recessions, with lasting consequences for worker earnings trajectories.

Quick Reference	
Key contribution	First quarterly J2J flow series from LEHD separating transition types by nonemployment duration; shows J2J flows lead the recession; documents procyclical earnings changes with widening nonemployment penalty.
Key weakness	Short P&P format (4 pages); no statistical inference; nine-state frame not nationally representative; cannot distinguish U from N.
Most interesting gap	The mechanism—workers not quitting vs. employers not hiring—is unresolved; identifying this distinction is crucial for policies targeting labour market fluidity.
Publishable extension	Construct Canadian J2J flow series from LEED 2000–2024; study resource- sector workers during commodity busts (Alberta 2014–16) as a quasi-experiment for sectoral J2J collapse and earnings losses.

Paper 7

Mueller (2017)

Separations, Sorting, and Cyclical Unemployment

Author: Andreas I. Mueller (Columbia GSB)

Journal: *American Economic Review* 107(7): 2081–2107

Year: 2017 **Field:** Labour economics; Macroeconomics

Classification: Empirical + structural (calibration)

7.1 Research Question

Does the composition of the pool of unemployed workers shift systematically over the business cycle, and in which direction—toward higher- or lower-wage workers? What drives this compositional shift—separations or job-finding rates? Can standard or modified theories of unemployment account for the observed pattern?

7.2 Motivation

It has long been known that the pool of *employed* workers shifts toward higher-wage workers in recessions (Solon, Barsky, Parker 1994), because low-wage workers are more likely to lose jobs. One might expect the *unemployed* pool to shift toward lower-wage workers. Mueller documents that this intuition is wrong: the unemployed pool shifts toward *high*-wage workers in recessions. The finding has profound implications for hiring incentives (compositional changes affect the average productivity of job applicants), for policy evaluation (ALMP participants are higher-quality in recessions), and for the unemployment volatility puzzle (Shimer 2005).

7.3 Main Contribution

1. **A new empirical fact:** The pre-displacement wage of unemployed workers rises strongly in recessions. A 1 pp increase in the unemployment rate is associated with a 1.5 pp increase in the average wage *rank* of the unemployed ($\rho = 0.67$, CPS ORG; 0.73, March CPS). This holds for raw wages, wage ranks, and Mincer residuals, and appears in every recession since 1962.
2. **Mechanism:** The shift is driven almost entirely by *separations*—high-wage workers have substantially more cyclical separation rates; job-finding rate cyclicalities are similar across wage groups.
3. **Theory evaluation:** Standard DMP models fail. Two extensions succeed: (i) lower variance of match productivity for high-ability workers; (ii) credit cash-flow constraints in recessions.

7.4 Relationship to the Literature

Extends Solon–Barsky–Parker (1994) to the unemployed pool, showing both pools can shift in the *same* direction (high-wage) in recessions. Challenges Pries (2008) who assumes the unemployed pool shifts toward low-ability workers. Adds a new challenge to the unemployment volatility puzzle (Shimer 2005): the high-ability unemployed raise vacancy returns, *dampening* the aggregate hiring response to adverse shocks.

7.5 Theoretical Framework and Economic Mechanism

Accounting Framework

The fraction of group i in the unemployed pool satisfies:

$$\phi_{it}^U = \phi_{it}^L \frac{U_{it}}{U_t}, \quad (7.1)$$

where ϕ_{it}^L is group i 's labour force share and U_{it} is group i 's unemployment rate. Using the Elsby–Michaels–Solon (2009) flow decomposition, compositional changes in the unemployed pool decompose (approximately) as:

$$d\phi_{it}^U \approx \phi_{it}^U (1 - \phi_{it}^U) \left[(1 - U_{it}^{ss})(d \ln s_{it} - d \ln f_{it}) - (1 - U_{jt}^{ss})(d \ln s_{jt} - d \ln f_{jt}) \right], \quad (7.2)$$

where s_{it} and f_{it} are group- i separation and job-finding rates. This shows compositional changes are driven by the *relative* log changes in flow rates, weighted by steady-state employment rates.

Why Standard Models Fail

In the baseline DMP model, low-wage workers are more likely to be near the separation threshold, so an adverse aggregate shock pushes more low-wage workers into unemployment—opposite to the data.

Extension 1: Lower Match Variance for High-Ability Workers

If the variance of match-specific productivity shocks is lower for high-ability workers, the density of matches at the separation threshold is *higher* for high-ability workers (their matches are more tightly clustered near the threshold). Aggregate shocks that shift the threshold cause a larger mass of high-ability matches to cross it, generating higher cyclicalities of high-wage separations. Critically, this calibration also preserves near-equal cyclicalities of job-finding rates across groups.

Extension 2: Credit Cash-Flow Constraints

Firms facing cash-flow constraints in recessions may separate from workers whose current match productivity is negative even if the NPV of the match is positive. Marginal matches with high-ability workers generate more negative current cash flows (workers paid above current productivity in anticipation of future returns), making them more sensitive to credit tightening.

7.6 Data

- **CPS Outgoing Rotation Groups (ORG):** Private-sector workers aged 16–64, 1980–2012; $N = 1,203,543$; 79,463 experienced ≥ 1 month of unemployment in interview months 5–8.
- **CPS March Supplement:** 1962–2012 (backward-looking wages; full 50-year coverage; no attrition issues).
- **Monthly CPS:** 1978–2012 (demographic composition of unemployed; no wage data).
- **NLSY79:** 1979–2012; $N = 6,923$; 193,467 yearly labour force status observations; used to distinguish permanent vs. transitory wage components.

7.7 Empirical Strategy

1. Compute average log previous wage, wage rank (ordinal), and Mincer residual for the unemployed pool in each year. HP-filter ($\lambda = 100$) and correlate with the HP-filtered aggregate unemployment rate.
2. Decompose compositional shifts into components by predicted wage (experience, education, gender, marital status, race, state, industry, occupation) using OLS regressions (Table 1).
3. Estimate group-specific (s_{it}, f_{it}) from matched CPS panels and apply the decomposition in equation (7.2) to isolate the separation vs. job-finding margin.
4. Calibrate baseline and extended DMP models to match average separation rates by wage group; compare predicted compositional shifts to data.

7.8 Identification Strategy

The main empirical claim is correlational, not causal. The paper documents a robust pattern across multiple data sources and specifications. The Mincer residual specification (controlling for demographics, industry, occupation, year) attempts to isolate the residual component. The NLSY79 analysis provides evidence that compositional changes are associated with *permanent* worker fixed effects rather than transitory match quality effects.

Key identification challenge: the composition of workers who lose their jobs could reflect true changes in separation behaviour *or* changes in the composition of at-risk workers. The paper addresses this through the wage rank specification (immune to wage compression) and the Mincer residual (controlling for observables), but cannot fully rule out unobservable worker selection into high-wage jobs.

7.9 Main Results

New Empirical Fact (Figures 3–4, Table 1)

Wage measure	CPS ORG	March CPS
ρ : avg log previous wage $\times$ unemp. rate	0.60	0.59
ρ : avg wage <i>rank</i> $\times$ unemp. rate	0.67	0.73
ρ : avg Mincer residual $\times$ unemp. rate	0.44	0.60
1 pp $\uparrow$ unemp. rate $\Rightarrow \Delta$ avg log wage	+2.77%	+2.59%
1 pp $\uparrow$ unemp. rate $\Rightarrow \Delta$ avg wage rank	+1.5 pp	+1.1 pp
1 pp $\uparrow$ unemp. rate $\Rightarrow \Delta$ avg Mincer residual	+0.75%	+0.95%

The pattern appears in *every recession since 1962* and is robust to all specifications.

Observable Drivers (Table 1)

All observable components shift in the same direction: the unemployed in recessions are more experienced, more educated, more likely to be male, married, white, and from high-wage industries and occupations. Industry accounts for $\approx 25\%$ of total compositional shift.

Separation vs. Job-Finding

The compositional shift is *almost entirely driven by separations*. High-wage workers have substantially more cyclical log separation rates; the cyclicity of log job-finding rates is similar across wage groups.

Model Results

- **Standard DMP:** Generates shifts toward *low*-ability workers—wrong direction.
- **Extension 1 (lower match variance):** Generates shifts toward high-ability workers; correct direction for all moments.
- **Extension 2 (credit constraints):** Correct direction for recessions with financial stress; internally inconsistent for recessions without credit disruption.

7.10 Economic Magnitude and Interpretation

A 1 pp rise in unemployment shifts the average unemployed worker from (say) the 50th to the 51.5th percentile of the wage distribution. Over a typical recession (≈ 3 pp unemployment rise), the average unemployed worker is at about the 54.5th percentile. The estimated dampening of aggregate hiring responses due to this compositional shift is up to $3.6\times$ the steady-state elasticity (Online Appendix G.1)—a quantitatively important challenge for the unemployment volatility puzzle.

7.11 Mechanisms

The preferred explanation is **lower variance of match-specific productivity for high-ability workers**:

- High-ability workers have matches clustered more tightly near the separation threshold.
- Aggregate shocks that shift the threshold by a given amount cause a larger mass of high-ability matches to cross it.
- This generates higher cyclicalities of separations without asymmetric cyclicalities of job-finding rates.

The **credit constraint** mechanism is plausible but has internal consistency problems: it predicts no compositional shift in recessions without clear financial market disruption (e.g. 1970–71), but the data show the shift in every recession back to 1962.

7.12 Robustness Checks

- Three CPS data sources (ORG, March, monthly) and NLSY79 give consistent results.
- Robust to: wage rank (immune to wage compression); Mincer residual (controls for observables); exclusion of long-term unemployed.
- NLSY79 distinguishes permanent worker effects from transitory match effects: shift is associated with permanent quality.
- Pattern holds in every recession 1962–2012.

7.13 Limitations

1. **Long-term unemployment exclusion:** CPS ORG excludes >12 months unemployed; becomes substantial post-2008 (up to 31.2%).
2. **Indirect mechanism test:** Cannot directly test whether match productivity variance differs across worker types—a maintained calibration assumption.
3. **Credit constraint test indirect:** Chodorow-Reich (2014)-based test is promising but not definitive.

7.14 Policy Implications

- **ALMP evaluation:** Higher-ability unemployed in recessions $\Rightarrow$ programmes more effective in recessions (Card, Kluve, Weber 2015).
- **UI design:** Cyclical shift toward high-wage unemployed means UI benefits (fixed fraction of prior wage) are *larger in absolute terms* during recessions—an automatic stabiliser amplification.
- **Hiring incentives:** High-ability unemployed raise returns to vacancy posting $\Rightarrow$ partial automatic stabiliser on the hiring side (dampens vacancy collapse).

7.15 Overall Assessment

Mueller (2017) is a clean, careful, and important empirical paper establishing a new stylised fact with exemplary robustness: three data sources, 50 years of data, multiple specifications, and a clear accounting framework. The theory section is more provisional but usefully narrows the space of viable explanations. The most important contribution is the implication for the unemployment volatility puzzle: compositional shifts toward high-ability workers in recessions *dampen* aggregate hiring responses.

Quality: Very good. Important new fact, careful empirics, sound theoretical interpretation.

7.16 Relevance to My Research

Relevant for any RDC project studying worker heterogeneity in unemployment dynamics. For Canadian work: LEED or matched LFS-style data could be used to test whether the Canadian unemployed pool also shifts toward high-wage workers in recessions, and whether this effect is amplified in resource provinces during commodity busts (where separated workers are predominantly from high-wage extraction jobs).

7.17 Potential Research Extensions

- Test the Mueller fact for Canada using matched LFS data; does the Canadian unemployed pool shift toward high-wage workers, and is this amplified in resource provinces during commodity busts?
- COVID-19 (primarily destroying low-wage service jobs) likely produced the *opposite* compositional shift—a natural out-of-sample test of the mechanism’s asymmetry.

7.18 Research Gaps

- Cannot distinguish the credit constraint from the lower match-variance mechanism; a firm-level test using credit supply shocks (e.g. bank failures) would be needed.
- COVID-19 provides an obvious out-of-sample test of whether the Mueller mechanism operates symmetrically—the composition of unemployment should shift toward *low*-wage workers.

7.19 Methodological Lessons

1. Equation (7.1) is essential: changes in the share of a group among the unemployed depend on *relative* unemployment rate changes (not absolute levels), so the higher-volatility unemployment of low-wage workers does *not* imply the unemployed pool shifts toward low-wage workers.
2. Wage *ranks* (ordinal) are more informative than levels because they are immune to wage compression as an alternative explanation.
3. The Elsbey–Michaels–Solon flow decomposition (equation 7.2) cleanly attributes compositional shifts to separations vs. job-finding, using log rates (not levels)

to account for different average flow rates across groups.

Five-Sentence Summary

Mueller (2017) documents a new fact using 50 years of CPS data: in recessions, the pool of unemployed shifts toward workers with *higher* pre-displacement wages, with a 1 percentage-point rise in unemployment associated with a 1.5 pp increase in the average wage rank of the unemployed—a pattern that holds in every recession since 1962, for raw wages, wage ranks, and Mincer residuals. The compositional shift is almost entirely driven by *separations*: high-wage workers have substantially more cyclical separation rates, while job-finding rate cyclicalities are similar across wage groups. Standard DMP models generate shifts in the *opposite* direction; two extensions succeed—lower variance of match productivity for high-ability workers and credit cash-flow constraints—with the former being the more broadly applicable explanation. The finding implies that the unemployed pool is higher quality in recessions, raising returns to vacancy posting and potentially *dampening* the aggregate hiring response to adverse shocks—an additional challenge for the unemployment volatility puzzle. Compositional changes in the unemployed pool are substantially larger in magnitude than the corresponding changes in the employed pool, implying standard composition-bias corrections are insufficient for statistics related to the unemployed.

Quick Reference

Key contribution	New empirical fact (robust across 3 data sources, 50 years): unemployed pool shifts toward high-wage workers in recessions, driven almost entirely by the higher cyclicalities of separations for high-wage workers.
Key weakness	The two successful theoretical extensions cannot be empirically distinguished; the credit constraint mechanism is internally inconsistent for recessions without financial stress.
Most interesting gap	COVID-19 likely produced the <i>opposite</i> compositional shift (low-wage service workers primarily displaced)—a natural out-of-sample test of whether the Mueller mechanism is asymmetric.
Publishable extension	Replicate Mueller’s decomposition for Canada using matched LFS data; test whether the shift toward high-wage unemployed is amplified in resource provinces during commodity busts, and assess whether it dampens hiring responses in those regions.

Paper 8

Krusell, Mukoyama, Rogerson & Şahin (2017)

Gross Worker Flows over the Business Cycle

Authors: Per Krusell, Toshihiko Mukoyama, Richard Rogerson, Ayşegül Şahin

Journal: *American Economic Review* 107(11): 3447–3476

Year: 2017 **Field:** Macroeconomics; Labour economics

Classification: Structural (heterogeneous-agent calibrated model)

8.1 Research Question

Can a parsimonious hybrid model—combining heterogeneous-agent labour supply with labour market frictions—simultaneously account for the cyclical properties of all six gross worker flow rates across employment (E), unemployment (U), and nonparticipation (N)?

8.2 Motivation

The E–U and U–E flows between employment and unemployment have been extensively studied (Blanchard–Diamond 1990; Shimer 2012). However, flows involving nonparticipation (N)—a state comprising $\approx 35\%$ of the population—were largely ignored, even though the flows into and out of N are as volatile as E–U flows. Two key counterintuitive facts motivate the paper: (i) the U-to-N transition rate (f_{UN}) is *procyclical*—workers move from unemployment to nonparticipation *more* in booms despite the participation rate being procyclical; (ii) both pure labour supply models (no U state) and pure search models (no participation margin) fail to capture this pattern. The paper builds a hybrid that accounts for all six flows simultaneously.

8.3 Main Contribution

1. **Documented new facts:** All six f_{ij} transition rates tabulated with two misclassification corrections (Abowd–Zellner; deNUNification). Key counterintuitive fact: f_{UN} is *procyclical*.
2. **First model to match all six steady-state flow rates** within 95% confidence intervals.
3. **Intertemporal substitution in a frictional model:** Even with a constant wage per efficiency unit, intertemporal labour supply effects emerge through the job ladder (higher f_{EE} in booms accelerates career advancement).

4. **Participation margin is first-order:** Flows between N and the other states account for $\approx 33\%$ of cyclical fluctuations in the unemployment rate.

8.4 Relationship to the Literature

Bridges three literatures: (i) heterogeneous-agent labour supply (Lucas–Rapping 1969; Chang–Kim 2006); (ii) search-matching (Mortensen–Pissarides 1994; Shimer 2012); (iii) gross worker flows with participation (Elsby, Hobijn, Şahin 2015; Shimer 2013). Distinguished from existing hybrid models by focusing on *gross worker flows* (not stocks alone) and by introducing UI, on-the-job search, and heterogeneous match quality for realistic calibration.

8.5 Theoretical Framework and Economic Mechanism

Individual Preferences

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^t [\log(c_t) - \alpha e_t - \gamma s_t], \quad (8.1)$$

where $c_t \geq 0$ is consumption, $e_t \in \{0, 1\}$ employment status, $s_t \in \{0, 1\}$ active search indicator, $\alpha > 0$ (disutility of work), $\gamma > 0$ (disutility of search).

Idiosyncratic Productivity and Match Quality

$\log z_{t+1} = \rho_z \log z_t + \varepsilon_{t+1}$ with $\varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$.

Each employment opportunity comes with an i.i.d. match quality draw $q \sim \text{LogNormal}(0, \sigma_q^2)$; employed workers can accept better matches via on-the-job search (λ_e).

Labour Market Frictions

Four friction parameters characterise the aggregate environment $\Lambda = (w, r, \lambda_u, \lambda_n, \lambda_e, \sigma)$:

- λ_u : offer arrival rate for active searchers (unemployed)
- λ_n : offer arrival rate for passive searchers (out of LF)
- λ_e : outside offer rate for employed workers
- σ : job separation rate

Bellman Equations (key)

For a worker with an employment opportunity (value W if accepted, J if rejected):

$$\begin{aligned} W(a, z, q, I^B, \Lambda) = \max_{c \geq 0, a' \geq 0} & \left\{ \ln c - \alpha \right. \\ & + \beta \mathbb{E} \left[(1 - \sigma - \lambda_e) V(a', z', q, \gamma', 0, \Lambda') + \lambda_e V(a', z', \max\{q, q'\}, \gamma', 0, \Lambda') \right. \\ & \left. \left. + \sigma(1 - \lambda_u) J(a', z', \gamma', 1, \Lambda') + \sigma \lambda_u V(a', z', q', \gamma', 1, \Lambda') \right] \right\} \quad (8.2) \end{aligned}$$

subject to $c + a' = (1 + r)a + (1 - \tau)wzq + T$.

For an unemployed worker who chooses active search:

$$U(a, z, \gamma, I^B, \Lambda) = \max_{c, a'} \left\{ \ln c - \gamma + \beta \mathbb{E} [\lambda_u V + (1 - \lambda_u) J] \right\} \quad (8.3)$$

subject to $c + a' = (1 + r)a + (1 - \tau)I^B b(z) + T$.

Mechanism for Procyclical f_{UN}

In booms, strongly attached workers find jobs quickly, leaving the unemployment pool populated by workers near the U–N indifference margin. These workers are more likely to transition to N, generating the *counterintuitive* procyclical f_{UN} .

8.6 Data

- **CPS gross worker flows:** Monthly CPS matched surveys, 1978:Q1–2012:Q3; six f_{ij} rates computed (Table 2).
- **Misclassification corrections:** (1) Abowd–Zellner (1985) reinterview-based; (2) deNUNification (recodes U→N→U sequences as N). Both substantially reduce flows involving N.
- **SIPP:** Validates model predictions for flow rates conditional on wealth.
- **Calibration targets:** LFPR (0.66), unemployment rate (0.068), f_{EU} (0.014), f_{NE} (0.022); J2J rate (2.2%/month); avg wage gain on J2J (3.3%).

8.7 Empirical Strategy

1. Tabulate cyclical properties of all six gross flow rates using HP-filtered ($\lambda = 1600$) quarterly series (Table 3).
2. Calibrate the model's steady state to match average flow rates (Table 4–5).
3. Subject calibrated model to aggregate shocks (drawn from cyclical variation in job-finding and separation rates); compare model-generated cyclical moments to data (Table 6).

8.8 Identification Strategy

Calibration, not econometric identification. Parameters are set through: (a) external values ($\rho_z = 0.996$, $\sigma_\varepsilon = 0.096$ from wage panel studies; $\tau = 0.30$; UI parameters from programme data); (b) steady-state moment matching (Table 4); (c) model fixed-point conditions. Aggregate shocks are measured from data on cyclical variation in job-finding and separation rates.

8.9 Main Results

Key Cyclical Facts (Table 3, AZ-corrected)

Flow	Std dev	$\rho(x, Y)$	Pattern
f_{EU}	0.089	-0.63	Countercyclical
f_{EN}	0.083	+0.43	Procyclical (counterintuitive)
f_{UE}	0.088	+0.76	
f_{UN}	0.106	+0.61	Procyclical (counterintuitive)
f_{NE}	0.103	+0.52	Procyclical
f_{NU}	0.072	-0.23	Weakly countercyclical

Note: f_{UN} procyclical means workers leave unemployment for N *more* in booms—counterintuitive from a discouraged-worker perspective.

Calibration Achievement (Table 5)

The model matches all six average flow rates within 95% CIs—the first structural model to achieve this. Previous models could not match N-adjacent flows.

Cyclical Performance (Table 6)

Model does a good job matching most cyclical moments; flows between participation and nonparticipation account for $\approx 33\%$ of unemployment rate fluctuations.

8.10 Economic Magnitude and Interpretation

The participation margin is far more important for cyclical unemployment than previously recognised: one-third of the rise in the unemployment rate during recessions reflects unemployed workers failing to exit to employment ($\downarrow f_{UE}$, $\uparrow f_{EU}$), but also the participation margin (f_{UN} , f_{NE}). The procyclical f_{UN} implies the standard “discouraged worker” narrative (workers give up searching in recessions) is a level effect, not a rate effect: the stock of discouraged workers rises in recessions not because the rate f_{UN} increases but because the stock of unemployed is larger.

8.11 Mechanisms

Two forces nearly cancel to produce the *weak* procyclicity of the participation rate:

- **Wealth effect:** Higher assets in booms reduce desire to work (reduces participation).
- **Intertemporal substitution:** Higher f_{EE} (J2J) in booms accelerates movement up the job-quality ladder $\Rightarrow$ higher return to participation now (increases participation).

Near-cancellation $\Rightarrow$ LFPR fluctuates little ($\sigma = 0.0026$) compared to the unemployment rate ($\sigma = 0.117$).

Procyclical f_{UN} : in booms, the unemployment pool is disproportionately composed of workers close to the U–N indifference margin; these workers transition to N at a higher rate, generating the counterintuitive procyclicality.

8.12 Robustness Checks

- Two misclassification corrections (Abowd–Zellner and deNUNification) give quantitatively similar cyclical properties; both shown in Table 3.
- Model evaluated against both correction methods in Table 6.
- SIPP wealth-conditional flow data used to validate model heterogeneity predictions.
- An earlier GE version (Krusell et al. 2012) gives similar results, confirming the PE assumption is not driving findings.

8.13 Limitations

1. **Partial equilibrium:** Prices and aggregate frictions are exogenous; wages and vacancy posting not endogenised.
2. **Friction shocks only:** The source of cyclical friction variation is not connected to TFP, monetary policy, or aggregate demand.
3. **No age, gender, or education heterogeneity:** All empirically important for N dynamics, especially for participation decisions.
4. **Quantitative mismatch:** Table 6 shows the model does not fully replicate all six cyclical moments simultaneously.

8.14 Policy Implications

- Policies targeting unemployment (UI, ALMP) affect flows from N as well—the participation margin is a first-order policy variable.
- The intertemporal substitution channel implies aggregate demand stimulus that increases J2J flows also raises participation—a positive feedback loop.
- Procyclical f_{UN} : UI recipients are more likely to stop searching and exit to N in *booms* (after finding a job, they stop claiming), consistent with UI moral hazard being cyclically variable.

8.15 Overall Assessment

A technically accomplished paper that fills an important gap in macro-labour economics. The empirical documentation of all six gross flow rates with careful misclassification corrections is valuable in itself. The calibration achievement—matching all six average flow rates within CIs—is a genuine advance. The conceptual contributions (participation fluctuations are first-order; intertemporal substitution operates through the job ladder) are important and influential.

Quality: Very good. Strong data work, careful calibration, novel model insights.

8.16 Relevance to My Research

Directly relevant for RDC work on Canadian labour market flows. The E/U/N three-state framework is applicable to matched LFS data or LEED/LWF. Testing whether Canadian gross flows show the same counterintuitive cyclical patterns (f_{UN} procyclical; large N-adjacent volatility) would be a valuable contribution, especially distinguishing resource-intensive from manufacturing provinces.

8.17 Potential Research Extensions

- Implement the gross-flow framework for Canada using matched monthly LFS microdata; test whether the counterintuitive procyclical f_{UN} holds in Canada and differs across provinces.
- Extend the model with sectoral heterogeneity (resource vs. manufacturing vs. services) to understand how commodity cycles drive differential E/U/N flows in Canada.

8.18 Research Gaps

- The cyclical source of friction variation is unexplained; integrating a production/vacancy-posting side (Mortensen–Pissarides structure) would close this gap.
- No treatment of long-run participation trends (declining since 2000); a low-frequency extension would be valuable.

8.19 Methodological Lessons

1. The Abowd–Zellner misclassification correction is essential for flows involving N; spurious U–N and N–U flows dramatically inflate measured N-adjacent volatility.
2. Matching all six *average* flow rates (not just aggregate stocks) is a far more stringent model test; passing the flow test automatically passes the stock test, but not vice versa.
3. Labour market stocks are a less stringent check on models than gross flows—when gross flows are available, they should be used as the primary calibration target.

Five-Sentence Summary

Krusell, Mukoyama, Rogerson, and Şahin (2017) document that gross worker flows into and out of nonparticipation are as volatile over the business cycle as the much-studied employment–unemployment flows, and that the U-to-N transition rate is procyclical—counterintuitive but explained by the changing composition of the unemployment pool in booms. They build a parsimonious hybrid model combining heterogeneous-agent labour supply with search frictions, the first structural model to match all six average gross flow rates within their 95% confidence intervals. Despite a constant wage per efficiency unit, intertemporal substitution operates through the job-quality ladder—higher job-to-job flows in booms increase career returns, generating a procyclical incentive to participate. Flows between participation and nonparticipation

account for approximately one-third of cyclical fluctuations in the unemployment rate—far larger than previously recognised. The paper establishes both empirically and theoretically that the participation margin is a first-order dimension of aggregate labour market dynamics that cannot be omitted from business cycle models of unemployment.

Quick Reference

Key contribution	First model to match all six steady-state E/U/N gross flow rates within CIs; establishes the counterintuitive procyclicality of f_{UN} ; documents that the N margin accounts for $\approx 33\%$ of unemployment fluctuations.
Key weakness	Partial equilibrium; aggregate friction shocks taken as given rather than derived; model does not fully match all cyclical moments quantitatively; no demographic heterogeneity.
Most interesting gap	The source of cyclical variation in friction parameters is left unexplained; a production side with vacancy posting and TFP shocks would complete the model.
Publishable extension	Implement the gross-flow framework for Canada using matched monthly LFS data; test whether the counterintuitive procyclical f_{UN} holds in Canada and differs between resource-intensive and manufacturing provinces.

Paper 9

Carrillo-Tudela & Visschers (2023)

Unemployment and Endogenous Reallocation Over the Business Cycle

Authors: Carlos Carrillo-Tudela (Essex) & Ludo Visschers (Edinburgh/UC3M)

Journal: *Econometrica* 91(3): 1119–1153

Year: 2023 **Field:** Labour economics; Macroeconomics

Classification: Structural (multisector business cycle model + SMM)

9.1 Research Question

To what extent does the cyclical nature of occupational mobility among the unemployed shape the aggregate unemployment rate and its duration distribution? Is cyclical unemployment driven primarily by occupation-wide productivity differences (net reallocation) or by individual workers' changing career prospects within occupations (idiosyncratic career shocks)?

9.2 Motivation

On average, 44% of unemployed U.S. workers change major occupational group (MOG) at reemployment. Occupational movers take longer to find jobs, and in recessions their duration increases $\approx 35\%$ more than stayers'. Yet standard multisector models (Lucas–Prescott 1974; Lilien 1982) emphasise *net* reallocation across sectors as the main driver of unemployment fluctuations—and in such models, gross mobility is typically constant or countercyclical. Carrillo-Tudela and Visschers document that gross occupational mobility is *procyclical* while net mobility is *countercyclical*, and that the dominant component is *excess mobility* (moves that cancel out at the occupation level). This motivates a new theory where idiosyncratic career shocks, not occupation-wide differences, drive cyclical unemployment.

9.3 Main Contribution

1. New empirical facts (SIPP, corrected for miscoding):

- 44.5% of unemployed workers change MOG at reemployment.
- Gross mobility is *procyclical*; net mobility is *countercyclical*.
- 90% of gross mobility is excess mobility (procyclical).
- Gross mobility increases with unemployment duration (mobility- duration profile).

2. **Multisector business cycle model with idiosyncratic career shocks (z):** workers face both occupation-wide productivity differences (p_o) and idiosyncratic career shocks (z). A “wait zone” between the separation and reallocation cutoffs generates rest/wait unemployment.
3. **Central quantitative finding:** Idiosyncratic career shocks (z), not occupation-wide differences (p_o), are the dominant driver of cyclical unemployment fluctuations.

9.4 Relationship to the Literature

Challenges the Lucas–Prescott (1974)/Lilien (1982)/Rogerson (1987) multisector framework. Related to Alvarez–Shimer (2011) rest/search model (but extends to dynamic business cycle). Complements Pilossoph (2014) and Chodorow-Reich–Wieland (2020)—which also find muted effects of net reallocation—but those papers have countercyclical or acyclical gross mobility, inconsistent with the data. Distinguishes from Kambourov–Manovskii (2008) by focusing exclusively on workers who go through unemployment.

9.5 Theoretical Framework and Economic Mechanism

Environment

Discrete time. Workers differ in: idiosyncratic productivity z_t (Markov, common across occupations—“career match” or career prospect); occupation-specific human capital x_t (accumulates when employed via learning-by-doing; depreciates during unemployment). Occupation-wide productivity $p_{o,t}$ and aggregate productivity A_t follow separate Markov processes.

Value Function—Unemployed Worker

$$W^U(\omega) = b + \beta E_{\omega'} \left[\max_{\rho(\omega')} \left\{ \rho R(\omega') + (1 - \rho) \left[\lambda(\theta(\omega')) W^E(\omega') + (1 - \lambda(\theta(\omega'))) W^U(\omega') \right] \right\} \right], \quad (9.1)$$

where $\rho \in \{0, 1\}$ is the reallocation decision, $R(\omega)$ is the expected net value of searching across occupations, $\theta(\omega)$ is labour market tightness in submarket (z, x) of occupation o .

Separation and Reallocation Cutoffs

Within each occupation, the model yields two productivity thresholds:

$$\text{Separation cutoff } z_s : \quad W^E(\omega) < W^U(\omega) \Leftrightarrow z < z_s \quad (9.2)$$

$$\text{Reallocation cutoff } z_r : \quad R(\omega) > W^U(\omega) \Leftrightarrow z < z_r \quad (9.3)$$

The calibration finds $z_s > z_r$ —creating a **wait zone** (z_r, z_s) where workers who separate prefer to remain attached to their occupation rather than switch, because they expect their career prospect z to recover.

Cyclical Widening of the Wait Zone

In recessions, aggregate productivity A falls $\Rightarrow$ wages fall $\Rightarrow$ both z_s and z_r rise. Because z_s rises more than z_r , the gap $z_s - z_r$ *widens*, pulling more workers into the wait zone $\Rightarrow$ more rest/wait unemployment $\Rightarrow$ longer spells, disproportionately for movers.

Imperfect Directed Search Across Occupations

Workers allocate search effort $\{s_{\bar{o}}\}$ across remaining occupations with $\sum_{\bar{o} \in O-} s_{\bar{o}} = 1$. Concavity of the matching function creates a trade-off between targeting the best occupation and the total probability of receiving any offer. This generates *excess mobility*: workers switch occupations even without systematic occupation-wide differences, driven by idiosyncratic career prospects.

9.6 Data

- **SIPP:** 1984–2008 panels (covering 1983–2014); EUE (employment $\rightarrow$ unemployment $\rightarrow$ employment) spells; excludes self-employed, military, agriculture.
- **Miscoding correction:** Novel degarbling matrix Γ identified from the 1985 $\rightarrow$ 1986 SIPP interviewing method change (independent to dependent). True occupational stayers have a $\approx 20\%$ chance of appearing as movers. Assumptions: (A1) independent classification errors; (A2) detailed balance in miscoding; (A3) strict diagonal dominance.
- **Cross-validation:** PSID 1968–1997; CPS 1979–2019.
- **Occupational classification:** 21 major groups (2000 SOC); also task-based (NRC, RC, NRM, RM).
- **Calibration:** Simulated Method of Moments (SMM).

9.7 Empirical Strategy

1. **Classification error correction:** Degarble observed flows $M^I = \Gamma' M \Gamma$ to recover true flows $M = (\Gamma^{-1})' M^I \Gamma^{-1}$.
2. **Mobility-duration profile:** For duration x , compute the proportion of workers who changed MOG among those with spells $\geq x$ months (Figure 1).
3. **Cyclical regressions (Table I):** Regress (log) gross mobility on HP-filtered unemployment rate; report across SIPP (corrected/uncorrected), CPS, with/without controls.
4. **Net/excess decomposition:** Net mobility $= \frac{1}{2} \sum_i |E_{-i} U E_i - E_i U E_{-i}| / EUE$; excess = gross – net.
5. **Model calibration:** SMM on moments including gross mobility rate, mobility-duration slope, cyclical, unemployment volatility, duration distribution, Beveridge curve.

9.8 Identification Strategy

The causal claim—that idiosyncratic career shocks (z), not occupation-wide differences (p_o), drive cyclical unemployment—is identified through model counterfactuals (shutting off z vs. p_o shocks). The empirical identification rests on:

- Gross flows order-of-magnitude larger than net flows $\Rightarrow$ occupation-wide differences cannot account for the data.
- Procyclical gross mobility (not net) $\Rightarrow$ models with countercyclical net reallocation are inconsistent.
- SMM estimates identify the z shock process from the mobility-duration profile slope.

The miscoding correction identification relies on the detailed balance assumption (A2), tested and supported using SIPP–PSID–CPS data.

9.9 Main Results

Empirical Results

Statistic	Value
Gross occupational mobility at reemployment	44.5%
Mobility at ≥ 9 months of duration	54.6%
Average excess mobility (of gross)	90%
Average net mobility	4.3%
Cyclicalit of gross mobility (SIPP, corrected): elast.	−0.145
Duration gap (mover vs. stayer): expansion	0.4 months
Duration gap: recession	0.9 months

Net mobility is *countercyclical*; excess mobility is *procyclical* and drives the procyclicality of gross mobility.

Model Results

- Quantitatively matches: procyclical gross mobility, countercyclical net mobility, downward-sloping Beveridge curve, unemployment volatility, duration distribution, cyclicalit of job-finding and separation rates.
- In recessions: rest/wait unemployment becomes relatively more prominent; search unemployment dominates in expansions.
- **Counterfactual:** Shutting off z shocks eliminates most of the cyclical variation in unemployment; shutting off p_o differences has little effect.

9.10 Economic Magnitude and Interpretation

The duration gap between movers and stayers grows from 0.4 to 0.9 months from expansion to recession—a 40% contribution to the overall increase in average unemployment duration across the cycle. This means roughly two-fifths of the aggregate

duration increase is attributable to the worsening situation of occupational movers specifically. The finding that 90% of gross mobility is excess (cancelling out at the occupation level) implies that the standard narrative of “structural unemployment” driven by workers fleeing hard-hit sectors accounts for only about 10% of actual occupational transitions.

9.11 Mechanisms

The central mechanism is the **endogenous widening of the wait zone** in recessions:

1. $A \downarrow \Rightarrow$ wages fall $\Rightarrow$ both z_s and z_r rise.
2. z_s rises more than z_r (because option value of staying in the occupation is more affected by falling wages than the option value of reallocation, which provides a fresh z draw).
3. More workers have career prospects $z \in (z_r, z_s) \Rightarrow$ they separate but *wait* rather than switch.
4. Waiting motive is stronger for occupational movers (who must redraw z upon switching) $\Rightarrow$ their spells lengthen more in recessions.

The **excess mobility mechanism**: workers who draw low z from the ergodic distribution after separation find their occupation unattractive and search in other occupations. Since most eventually draw a new z (often returning to prior occupation), these moves cancel at the occupation level $\Rightarrow$ large gross but small net flows.

9.12 Robustness Checks

Extensive (Online Supplementary Material + Supplementary Appendices):

- Results hold under alternative classifications (21-group SOC, task-based NR-C/RC/NRM/RM).
- Procyclicality robust using SIPP (corrected/uncorrected), CPS 1979–2019, with/without demographic controls.
- Mobility-duration profile robust to nonemployment spells, alternative spell definitions, CPS 1994-redesign restriction.
- Classification error assumptions (A1–A3) tested with PSID and CPS.
- Non-targeted moment: repeat mobility (63.4% of stayers remain stayers; 54.4% of movers become movers again)—matched by the model.

9.13 Limitations

1. **EUE spells only**: Workers who never return to employment are excluded; may miss extreme reallocation cases.
2. **Coarse occupation classification**: 21 major groups may miss finer mobility patterns.
3. **BRE restriction**: Block Recursive Equilibrium restricts value functions to individual states plus aggregate—a technical tractability assumption.

4. **Exogenous wage determination:** Nash bargaining; richer wage dynamics (posting, rigidity) may alter quantitative results.
5. **US-specific calibration:** The z shock process and mobility rates are calibrated to U.S. data; generalisability unclear.

9.14 Policy Implications

- Idiosyncratic career shocks dominate $\Rightarrow$ standard *sectoral retraining* policies address only $\approx 10\%$ of occupational transitions (net flows). Most mobility is driven by individual career dynamics.
- The wait zone mechanism implies workers appearing “structurally unemployed” are often rationally waiting for career recovery. Policies forcing rapid reallocation (e.g. strict UI time limits) may push workers to accept worse occupational matches prematurely.
- Occupational movers have 35% longer duration increases in recessions $\Rightarrow$ supports differential UI policies for occupational switchers (longer or more generous benefits during transitions).

9.15 Overall Assessment

An exceptional paper combining careful empirical measurement (novel miscoding correction), a tractable theoretical model, and rigorous SMM calibration. The main empirical fact—procyclical gross mobility, countercyclical net mobility, excess mobility dominates—overturns a widely held prior. The model’s success in matching this complex pattern simultaneously with the cyclical properties of unemployment duration is impressive.

Quality: Outstanding. A landmark contribution to occupational mobility, unemployment dynamics, and business cycles.

9.16 Relevance to My Research

Highly relevant for RDC work. Canadian LEED/LEAP data may allow measurement of occupational mobility among the unemployed, testing whether the procyclical gross/countercyclical net finding holds in Canada. The wait-zone mechanism has direct relevance for resource-sector workers during commodity busts.

9.17 Potential Research Extensions

- Test the mobility-duration profile and its cyclicity for Canada using LWF/LEED data.
- Study whether the wait zone is wider for resource-sector workers in Canada during commodity busts (high occupation-specific human capital $\Rightarrow$ stronger waiting motive).

9.18 Research Gaps

- The paper does not model long-run secular changes in occupational mobility (automation, polarisation); the excess mobility mechanism might interact with

technological change.

- Firm actions (RD, capital investment) could endogenously affect workers' career prospects z —this feedback is absent.

9.19 Methodological Lessons

1. The degarbling matrix Γ identified from a data collection design change (SIPP interviewing method) is a powerful and broadly applicable tool for measuring occupational mobility free of coding error.
2. Decomposing gross mobility into net and excess components is essential: the two have completely different cyclical properties; aggregating them would obscure the key empirical patterns.
3. Block Recursive Equilibrium is an effective approach to tractability in rich multisector heterogeneous-agent models—it rules out the joint productivity distribution as a state variable while preserving full individual-level heterogeneity.

Five-Sentence Summary

Carrillo-Tudela and Visschers (2023) document, using SIPP data with a novel miscoding correction, that 44% of unemployed U.S. workers change major occupational group at reemployment, gross occupational mobility among the unemployed is procyclical, net mobility is countercyclical, and 90% of gross mobility is excess mobility—patterns inconsistent with standard multisector models. They develop a multisector business cycle model with idiosyncratic career shocks (z), occupation-specific human capital, and DMP matching frictions; within each occupation the separation cutoff exceeds the reallocation cutoff, creating a wait zone where separated workers prefer to remain attached to their occupation. In recessions, aggregate productivity falls, the wait zone widens endogenously, more workers enter rest/wait unemployment, and duration spells lengthen disproportionately for occupational movers—generating the observed 35% larger duration increase for movers than stayers in downturns. The model is calibrated by SMM and quantitatively matches the mobility-duration profile, procyclical gross mobility, countercyclical net mobility, unemployment volatility, and the Beveridge curve simultaneously. The core finding is that cyclical unemployment is driven by workers' changing idiosyncratic career prospects, not occupation-wide differences—implying standard sectoral retraining policies address only a small fraction ($\approx 10\%$) of actual occupational transitions.

Quick Reference

Key contribution	First model to simultaneously reproduce procyclical gross and countercyclical net occupational mobility among the unemployed; shows idiosyncratic career shocks (not occupation-wide differences) drive cyclical unemployment through an endogenous wait-zone mechanism.
Key weakness	EUE spells only (non-returners excluded); BRE restriction for tractability; exogenous Nash wage determination; US-specific calibration.
Most interesting gap	Interaction between automation/polarisation and the excess mobility mechanism: if technological change destroys career prospects within occupations, the wait zone should be wider and cyclical amplification larger.
Publishable extension	Test the mobility-duration profile and its cyclicity for Canada using LEED/LWF data; study whether resource-sector workers exhibit stronger wait-zone behaviour during commodity busts due to their high occupation-specific human capital.

Paper 10

Asplund & Nocke (2006)

Firm Turnover in Imperfectly Competitive Markets

Authors: Marcus Asplund (LBS/Columbia) & Volker Nocke (UPenn)

Journal: *Review of Economic Studies* 73(2): 295–327

Year: 2006 **Field:** Applied microeconomics; Industrial organisation

Classification: Structural (theoretical + empirical test)

10.1 Research Question

How do market size and fixed costs determine firm turnover (entry and exit rates) and the age distribution of firms in imperfectly competitive markets? Does the price-competition effect generated by larger markets systematically accelerate firm turnover by raising the exit threshold?

10.2 Motivation

Industries differ substantially in firm turnover and entry/exit rates are highly positively correlated across industries (Dunne, Roberts, Samuelson 1988). Existing theory (primarily Hopenhayn 1992) explains this in perfectly competitive markets: entry costs and fixed costs affect turnover by changing the exit threshold. However, in perfectly competitive models, market size has *no effect* on firm turnover because the entry cost is independent of the number of firms. In imperfectly competitive industries, a new mechanism operates: larger markets → more firms → lower price-cost margins → higher exit threshold → faster turnover. Asplund and Nocke formalise and test this **price-competition effect**.

10.3 Main Contribution

1. **Theoretical:** Shows (under Assumption 2: price competition effect) that firm turnover is *increasing in market size*—a novel prediction unavailable from Hopenhayn’s competitive benchmark.
2. **Empirical:** Tests the prediction using data on hair salons in Sweden across municipalities of different sizes. FOSD tests and regressions confirm that firms are younger in larger markets and in markets with higher fixed costs.
3. **Comparative dynamics:** Derives the full set of signed comparative dynamics results for entry costs, fixed costs, and market size.

10.4 Relationship to the Literature

Extends Hopenhayn (1992) to imperfect competition. Differs from Melitz (2003) in that Dixit-Stiglitz monopolistic competition *does not* satisfy Assumption 2 (constant markup $\Rightarrow$ no price competition effect; turnover is independent of market size in Melitz). Standard Cournot and linear demand models *do* satisfy Assumption 2. Assumption 2 is related to but finer than Boone’s (2000) “toughness of competition” analysis.

10.5 Theoretical Framework and Economic Mechanism

Model Structure

Discrete time, infinite horizon. Firm efficiency $c \in [0, 1]$ (lower c = more efficient). Efficiency evolves as a mixture: with probability α , c stays the same; with probability $1 - \alpha$, a new c is drawn from $G(\cdot)$. Sunk entry cost $\varepsilon > 0$; fixed per-period production cost $\phi > 0$.

Reduced-Form Profit Function

Firm gross profit (net of fixed costs) when the measure of active firms is μ :

$$S \pi(c; \mu) \geq 0, \quad (10.1)$$

where S is market size. Assumptions on π :

- **(MON)**: $\pi(\cdot; \mu)$ strictly decreasing in c on $[0, c(\mu))$; zero for $c \geq c(\mu)$.
- **(DOM)**: FOSD ordering of firm distributions is preserved by the profit ordering.
- **(ORD)**: The ordering $(M, \succeq)$ is complete.
- **(CON)**: $\pi(c; \mu)$ is continuous.
- **Assumption 1 (market share effect)**: For $\mu' \succ \mu$, $\pi(c; \mu) - \pi(c; \mu')$ is strictly decreasing in c —efficient firms suffer larger *absolute* profit reductions from intensified competition.
- **Assumption 2 (price competition effect)**: For $\mu' \succ \mu$, $\pi(c; \mu')/\pi(c; \mu)$ is strictly decreasing in c —less efficient firms suffer larger *percentage* profit reductions.

Assumption 2 is the key new ingredient. It holds for standard Cournot and linear demand models but *fails* for Dixit-Stiglitz (constant markup).

Stationary Equilibrium Value Function

$$V(c) = \max\{0, \bar{V}(c)\}, \quad \bar{V}(c) = [S\pi(c; \mu) - \phi] + \delta \left[\alpha V(c) + (1 - \alpha) \int_0^1 V(z) G(dz) \right]. \quad (10.2)$$

Entry and Exit Conditions

In stationary equilibrium with positive entry and exit:

$$(X) \quad S\pi(c^*; \mu) - \phi + \delta(1 - \alpha)\varepsilon = 0 \quad (\text{optimal exit at threshold } c^*) \quad (10.3)$$

$$(E) \quad \int_0^{c^*} [S\pi(c; \mu) - \phi + \delta(1 - \alpha)\varepsilon] G(dc) = (1 - \alpha\delta)\varepsilon \quad (\text{free entry}) \quad (10.4)$$

The firm at threshold c^* earns negative current net profit $[S\pi(c^*; \mu) - \phi < 0]$ but stays because of the option value of getting a new efficiency draw: $\delta(1 - \alpha)\varepsilon > 0$.

Central Comparative Dynamics Result (Proposition 4)

Under Assumption 2, an increase in market size S :

1. Raises c^* (higher exit threshold—less efficient firms exit).
2. Increases the rate of firm turnover.
3. Shifts the age distribution of firms toward younger firms in the sense of first-order stochastic dominance.

Mechanism: Larger $S \rightarrow$ more entrants $\rightarrow$ price-cost margins fall $\rightarrow$ percentage profit reduction is larger for inefficient firms (Assumption 2) $\rightarrow$ their value falls below the exit threshold $\rightarrow$ they exit $\rightarrow$ marginal surviving firm must be more efficient $\rightarrow c^*$ rises $\rightarrow$ faster turnover $\rightarrow$ younger age distribution.

10.6 Data

- **Swedish hair salons (1993–2001):** Age distribution across Swedish municipalities of varying population sizes.
- **Advantages:** (i) Single industry $\Rightarrow$ cross-industry shock volatility held constant. (ii) Local geographically-distinct markets. (iii) Fixed costs vary observably across municipalities.
- **Tests:** Non-parametric FOSD tests of age distributions; OLS and quantile regressions of firm age on log population and fixed cost proxies.

10.7 Empirical Strategy

1. For pairs of municipalities of different sizes, test whether the CDF of firm ages in the larger market is above (younger) the CDF in the smaller market (FOSD test).
2. Regress firm age on log population (market size) and observable fixed cost proxies in OLS and quantile regressions.
3. Exploit municipality-specific variation in fixed costs to test Proposition 5 (higher fixed costs $\Rightarrow$ younger firms).

10.8 Identification Strategy

The key identifying variation is *within-industry, cross-market variation in market size and fixed costs*. By studying a single industry across many geographically distinct local markets, the authors eliminate the main confound in cross-industry tests (cross-industry variation in shock volatility). The identifying assumption is that local market size (population) is exogenous to the age distribution of hair salons.

Limitation: No instrumental variable for market size; potential reverse causality if cities with younger firms attract more consumers.

10.9 Main Results

Theoretical Comparative Dynamics

Parameter change	Effect on c^*	Effect on turnover
$\uparrow$ Entry cost ε	$\downarrow$	$\downarrow$
$\uparrow$ Market size S	$\uparrow$	$\uparrow$ (Assumption 2)
$\uparrow$ Fixed cost ϕ	$\uparrow$	$\uparrow$ (Assumption 2)

Empirical Results (Hair Salons)

- Non-parametric FOSD tests: the age distribution in larger municipalities FOSD-dominates that in smaller municipalities—firms are *systematically younger* in larger markets. ✓
- OLS: log market size has a negative, statistically significant coefficient on average firm age. ✓
- Higher fixed cost proxies: associated with younger firms, consistent with Proposition 5. ✓
- Quantile regressions: the effect is pervasive across the full age distribution, not driven by outliers.

10.10 Economic Magnitude and Interpretation

An increase in market size of one standard deviation reduces average firm age by a meaningful amount. The finding that higher fixed costs are associated with *younger* firms (more turnover) is particularly counterintuitive: it implies that regulations or compliance costs that raise the per-period cost of operating do not necessarily reduce dynamism—they can *increase* turnover by forcing out low-productivity incumbents.

10.11 Mechanisms

The price-competition effect operates as:

1. Larger market $\rightarrow$ free entry drives in more firms.
2. Price-cost margins fall for all active firms (competition intensifies).

3. Assumption 2 (price effect): percentage decline in profits is larger for less efficient (higher c) firms, because price falls are proportionally more damaging when margins are already thin.
4. High-cost incumbents cross the exit threshold first $\rightarrow$ they exit.
5. Marginal surviving firm must be more efficient (c^* rises).
6. More efficient survivors have shorter expected lifetimes for any given entrant $\rightarrow$ higher turnover $\rightarrow$ younger age distribution.

10.12 Robustness Checks

- Main predictions verified under Cournot competition (Appendix); Assumptions 1 and 2 hold in this canonical setting.
- Results extended to the linear demand model with perceived quality differences (isomorphic representation).
- FOSD tests are non-parametric—no distributional assumptions.
- Quantile regressions confirm results not driven by outliers.
- Robustness to alternative fixed cost proxies.
- The Dixit-Stiglitz failure of Assumption 2 is explicitly demonstrated and discussed as a limitation for trade applications.

10.13 Limitations

1. **No IV for market size:** Potential reverse causality.
2. **Single industry:** Hair salons are low-tech/labour-intensive; generalisability to capital-intensive or high-tech sectors unclear.
3. **Dixit-Stiglitz incompatibility:** Most widely used trade model (Melitz 2003) does not satisfy Assumption 2.
4. **Stylised efficiency process:** Mixture of persistence and random redraw; more realistic AR(1) processes may yield different quantitative predictions.
5. **No worker side:** Welfare effects of higher turnover on workers are outside the model.

10.14 Policy Implications

- **Entry regulation:** Lower entry costs increase turnover and efficiency (as in Hopenhayn 1992).
- **Fixed production costs:** Higher fixed costs (from compliance burdens) increase turnover by forcing exit of less productive firms— a counterintuitive policy implication.
- **Economic integration:** Larger integrated markets (EU integration, free trade areas) generate more intense price competition, raise exit thresholds, and improve aggregate productivity—but at the cost of incumbent firm stability.

10.15 Overall Assessment

Asplund and Nocke (2006) is an elegant paper that cleanly identifies a new mechanism (the price-competition effect) not present in Hopenhayn's competitive benchmark and provides a consistent theory and empirical test. The within- industry cross-market design is imaginative and appropriate. The main weakness is the lack of a causal identification strategy for market size and the restriction to a single low-tech industry. The finding that Dixit-Stiglitz (most widely used competition model in trade) does not generate the price competition effect is an important limitation.

Quality: Good. Clean theory, appropriate empirical test, but narrow empirical scope.

10.16 Relevance to My Research

Relevant for research on firm dynamics in Canada at the RDC. Using LEAP or T2 corporate tax data, one could test whether firm turnover is higher in larger Canadian metropolitan areas (Toronto vs. Prince George), controlling for industry, consistent with the market size $\rightarrow$ turnover prediction.

10.17 Potential Research Extensions

- Test the market size $\rightarrow$ turnover prediction for Canadian service industries (restaurants, retail) using LEAP/LEED data across municipalities.
- Examine whether CETA/CPTPP increased firm turnover in Canadian import-competing industries through the price-competition channel.

10.18 Research Gaps

- The aggregate productivity consequences of market-size-driven turnover are not quantified—a Foster-Haltiwanger-Krizan (2001) decomposition of productivity growth across markets of different sizes would be the natural next step.
- No worker side: welfare effects of higher firm turnover on workers (displacement, wage changes) are outside the model.

10.19 Methodological Lessons

1. Testing theory within a *single industry* across geographic markets is a powerful design eliminating the main confound (cross-industry shock volatility) in cross-industry tests.
2. Non-parametric FOSD tests of age distributions are more convincing than tests based on a single summary statistic (mean age), as they test the full distributional prediction.
3. The Assumption 1/2 taxonomy (market share effect vs. price competition effect) is a useful organising framework for comparing models of imperfect competition on their firm dynamics implications.

Five-Sentence Summary

Asplund and Nocke (2006) develop a stochastic dynamic model of a monopolistically competitive industry showing that firm turnover is increasing in market size through a price-competition effect: larger markets support more firms, which intensifies competition, and the percentage profit reduction is larger for less efficient (higher-cost) firms (Assumption 2), raising the exit threshold and accelerating turnover. This prediction—absent from Hopenhayn’s (1992) competitive benchmark and inconsistent with the Dixit-Stiglitz model—is tested using data on Swedish hair salons across municipalities of varying sizes, where within-industry geographic variation eliminates the main confound in cross-industry tests. Non-parametric FOSD tests and OLS regressions confirm that larger markets have age distributions shifted toward younger firms, and markets with higher fixed costs (Proposition 5) similarly have younger firms. The model delivers clean comparative dynamics: higher entry costs reduce turnover by protecting incumbents, while higher fixed production costs increase turnover through a selection effect. The paper establishes that market size is a first-order determinant of firm dynamics under imperfect competition— a mechanism missing from competitive industry models and from the widely used Dixit-Stiglitz/Melitz trade framework.

Quick Reference

Key contribution	Novel theoretical prediction that firm turnover is increasing in market size under imperfect competition (price-competition effect); first paper to embed this in a stochastic dynamic equilibrium and test it empirically with a within-industry cross-market design.
Key weakness	No IV for market size (potential reverse causality); limited to a single low-tech industry (hair salons in Sweden); central result does not hold under Dixit-Stiglitz (most common trade model).
Most interesting gap	The aggregate productivity implications of market-size-driven turnover are not quantified; a Foster-Haltiwanger-Krizan (2001) decomposition of productivity growth across markets of different sizes would complete the welfare analysis.
Publishable extension	Test the Asplund-Nocke market size $\rightarrow$ firm turnover prediction for Canadian service industries using LEAP data across Census Metropolitan Areas; study whether CETA/CPTPP increased turnover in import-competing industries through the price-competition channel.

Paper 11

Acemoglu, Akcigit, Alp, Bloom & Kerr (2018)

Innovation, Reallocation, and Growth

Authors: Daron Acemoglu, Ufuk Akcigit, Harun Alp, Nicholas Bloom, William Kerr

Journal: *American Economic Review* 108(11): 3450–3491

Year: 2018 **Field:** Macroeconomics; Endogenous growth; Applied microeconomics

Classification: Structural (endogenous growth model + SMM estimation)

11.1 Research Question

What are the implications of different types of industrial policies—R&D subsidies, operating cost subsidies, and entry subsidies—for innovation, reallocation, and growth in an economy where firms are heterogeneous in their innovative capacity? Can *taxing* the continued operation of incumbents improve welfare by freeing skilled labour for R&D by high-type firms?

11.2 Motivation

Industrial policies subsidising incumbent firms are pervasive worldwide (EU: EUR 1.18 trillion in 2010, or 9.6% of GDP). Yet existing models either lack endogenous exit (and cannot model exit-targeting policies) or lack firm heterogeneity in innovative capacity (and cannot model R&D input misallocation). The paper fills this gap: it shows the key market failure is the misallocation of *R&D inputs* (skilled labour) between high- and low-type firms, and demonstrates that operation taxes—not R&D subsidies—are the most effective instrument for positive selection.

11.3 Main Contribution

1. **Model:** A quality-ladder endogenous growth model with two-type firm heterogeneity ($\theta_h > \theta_l$), endogenous exit via obsolescence, creative destruction, and type transitions (high→low at rate ν). Franchise values solved in closed form.
2. **Structural estimation:** 8 parameters estimated by SMM targeting 18 moments from U.S. Census microdata on innovative manufacturing firms (LBD, CMF, NSF R&D, NBER Patents), 1987–1997. Covers 98% of U.S. industrial R&D.

3. **Policy analysis (Table 12):** Incumbent R&D subsidy = +0.63% welfare; operation subsidy = -0.20%; entry subsidy = -1.64%; optimal operation *tax* (69%) = +1.40%; social planner = +4.47%.

11.4 Relationship to the Literature

Extends Klette–Kortum (2004) and Lentz–Mortensen (2008) by adding: (i) endogenous exit via obsolescence; (ii) two-type firm heterogeneity with high-to-low transitions; (iii) fixed costs of operation. Without (i) there is no exit margin to tax; without (ii) all firms are homogeneous and R&D subsidies are not differentially distortionary. Also relates to the misallocation literature (Hsieh–Klenow 2009; Restuccia–Rogerson 2008), but focuses on misallocation of *R&D inputs* rather than production inputs.

11.5 Theoretical Framework and Economic Mechanism

Preferences and Technology

CRRA preferences over a CES consumption aggregate:

$$U_0 = \int_0^\infty e^{-\rho t} \frac{C(t)^{1-\vartheta} - 1}{1-\vartheta} dt, \quad C(t) = \left(\int_{\mathcal{J}(t)} c_j(t)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}. \quad (11.1)$$

Product j is produced by its leading-edge monopolist at cost $w_u/\hat{q}_j$ ($\hat{q}_j \equiv q_j/w_u$), generating equilibrium profit:

$$\pi(\hat{q}_j) = \Pi \hat{q}_j^{\varepsilon-1}, \quad \Pi \equiv \frac{1}{\varepsilon-1} \left(\frac{\varepsilon-1}{\varepsilon} \right)^\varepsilon. \quad (11.2)$$

Firm Type Dynamics

Entrants draw $\theta = \theta_h$ with probability α and $\theta = \theta_l$ with probability $1-\alpha$. High-type firms transition to low-type at rate $\nu > 0$; low-type is absorbing. A type- k firm with n product lines and h skilled R&D workers innovates at flow rate:

$$X_f = \theta_k^\gamma n^\gamma h^{1-\gamma}, \quad C(x, n, \theta) = w_s n G(x, \theta), \quad G(x, \theta) = x^{\frac{1}{1-\gamma}} \theta^{-\frac{\gamma}{1-\gamma}}. \quad (11.3)$$

Value Functions (Franchise Value Additivity)

Lemma 1: The firm value is additive across product lines: $\tilde{V}_k(\hat{\mathcal{Q}}) = \sum_{\hat{q} \in \hat{\mathcal{Q}}} \Upsilon_k(\hat{q})$, where the franchise value $\Upsilon_k(\hat{q})$ satisfies the ODE (Lemma 2):

$$(r + \tau + \phi + (\varepsilon - 1)g) \Upsilon_l(\hat{q}) - \frac{\partial \Upsilon_l}{\partial \hat{q}} \frac{\partial \hat{q}}{\partial w_u} \dot{w}_u = \Pi \hat{q}^{\varepsilon-1} - \tilde{w}_s \phi + \Omega_l, \quad \hat{q} > \hat{q}_{l,\min}, \quad (11.4)$$

with closed-form solution (Proposition 1). The exit threshold is:

$$\hat{q}_{k,\min} = \left(\frac{\tilde{w}_s \phi - \Omega_k}{\Pi} \right)^{\frac{1}{\varepsilon-1}}, \quad k \in \{h, l\}. \quad (11.5)$$

Growth and Key Market Failure

Aggregate growth equals $g = \lambda\tau$ (Proposition 2; standard quality ladder). The key market failure is the underinvestment in R&D relative to the social optimum: because future innovators build on current quality ($q_j(t+) = q_j + \lambda\bar{q}$), current innovation creates a positive spillover. This underinvestment is especially costly from the perspective of high-type firms, whose marginal product of R&D is higher. The social planner wishes to shift skilled labour from low-type firm operations to high-type R&D.

Why operation taxes work and R&D subsidies fail:

- Low-type firms are disproportionately near the obsolescence threshold $\hat{q}_{l,\min}$ (Figure 2 in the paper).
- An operation tax raises $\hat{q}_{k,\min}$, forcing exit of low-type product lines near the margin. This frees skilled labour for R&D by high-type firms, generating strong positive selection.
- R&D subsidies raise demand for skilled labour from *both* types, raising the skilled wage and crowding out entrant R&D without improving the composition of active firms.

11.6 Data

- **LBD:** Annual employment for all private-sector establishments, 1976–present.
- **CMF:** Quinquennial detailed plant/firm records; years 1987, 1992, 1997.
- **NSF R&D Survey (RAD):** Annual/biannual firm-level R&D expenditures; certainty threshold > \$1m R&D.
- **NBER Patent Database:** USPTO patents matched to LBD by name/location.
- **Sample:** “Continuously innovative” firms (R&D or patenting in every CMF period). 17,055 obs. from 9,835 firms; 98% of U.S. industrial R&D; 50% of manufacturing employment.

11.7 Empirical Strategy

SMM targeting 18 moments (firm exit rates by age/size, size transition rates, employment and sales growth by age/size, R&D intensity, 5-year entrant employment share, fixed cost/R&D labour ratio, aggregate growth). Aggregate growth is weighted 5× the micro-moments. Bootstrap standard errors (1,000 replications, stratified by firm age, size, year, industry).

Computational Algorithm

Solve the stationary equilibrium as a fixed point of six aggregate variables:

$$\{\tilde{w}_s, \Phi_h, \Phi_l, \hat{q}^{\text{avg}}, \mathbb{E}[\Upsilon_h(\hat{q} + \lambda\hat{q}^{\text{avg}})], \mathbb{E}[\Upsilon_l(\hat{q} + \lambda\hat{q}^{\text{avg}})]\}. \quad (11.6)$$

2^{17} firms simulated for 10,000 periods (each period = 0.02 years) to generate model-implied moments.

11.8 Identification Strategy

8 parameters are estimated from 18 moments (overidentified by 10). Each parameter is primarily identified by specific moments:

- ϕ (fixed cost): ratio of fixed-cost to R&D workers.
- α, ν : age gradient in exit rates and R&D intensity (most young firms are high-type; transitions to low-type raise exit rates for older cohorts).
- λ : aggregate growth rate.
- θ_h/θ_l : ratio of R&D intensity between large-old and small-young firms.
- ϕ (exogenous exit): exit rates of large-old firms (unlikely to exit endogenously).

11.9 Main Results

Parameter Estimates (Table 2)

Parameter	Description	Value
ϕ	Fixed cost of operation	0.216
θ_h	High-type innovative capacity	1.751
θ_l	Low-type innovative capacity	1.391
θ_E	Entrant innovative capacity	0.024
α	P(high-type entrant)	0.926
ν	High→low transition rate	0.206/year
λ	Innovation step size	0.132
ϕ (exog. exit)	Exogenous destruction rate	0.037

Key implications: high-type firms are 26% more innovative than low-type ($\theta_h/\theta_l \approx 1.26$); 93% of entrants start as high-type; 21%/year probability of transitioning to low-type — strong negative selection as firms age.

Baseline Economy (Table 4)

Equilibrium growth: $g = 2.26\%$; creative destruction: $\tau = 17.2\%$; R&D employment share: $L_{R\&D}/L_S = 19.9\%$; product line shares: $\Phi_h = 6.3\%$, $\Phi_l = 55.0\%$, $\Phi_{np} = 38.7\%$ (inactive).

Policy Experiments (Table 12)

Policy	Growth	Welfare
Baseline	2.26%	100.00
Incumbent R&D subsidy (14%, 1% GDP)	2.34%	100.63
Operation subsidy (4%, 1% GDP)	2.24%	99.80
Entry subsidy (65%, 1% GDP)	2.25%	98.36
Optimal operation tax (−69%)	2.54%	101.40
Social planner	2.94%	104.47

Social planner raises low-type exit threshold ($\hat{q}_{l,\min}$: 1.47 $\rightarrow$ 2.40), lowers high-type threshold ($\hat{q}_{h,\min}$: 1.30 $\rightarrow$ 0.28), and raises Φ_h/Φ_l from 0.11 to 7.93.

11.10 Economic Magnitude and Interpretation

The welfare gain from the optimal operation tax (+1.40%) is 2.2 $\times$ that of the incumbent R&D subsidy (+0.63%), despite both costing 1% of GDP. The social planner achieves a 4.47% welfare gain and raises growth by 0.68 pp (2.26 $\rightarrow$ 2.94%), showing the equilibrium is substantially inefficient. The operation tax is “the most effective implementable policy”: by targeting firms near the exit margin—disproportionately low-type—it achieves selection effects that uniform R&D subsidies cannot.

11.11 Mechanisms

The central mechanism operates through **reallocation of skilled labour**:

1. Too many skilled workers cover fixed costs of low-type firm operations; too few perform R&D.
2. Operation tax raises $\hat{q}_{l,\min} \Rightarrow$ forces exit of low-type product lines near the margin $\Rightarrow$ frees skilled labour.
3. Freed skilled labour flows to R&D by high-type firms (higher marginal product) $\Rightarrow$ positive selection: Φ_h/Φ_l rises.
4. R&D subsidies fail because they increase skilled labour demand from *both* types, raising the skilled wage and crowding out entrant R&D.

11.12 Robustness Checks

- Results replicated with employment-weighted moments (vs. unweighted baseline); including non-innovative firms (dropping R&D moments); excluding mergers and acquisitions.
- Model variants: fixed costs using skilled + unskilled labour; factor reallocation costs; more than two firm types; endogenous skill supply.
- Non-targeted moments: product line distribution (CMF Product Trailers), growth persistence by firm type, management scores from MOPS all closely matched.

11.13 Limitations

1. Sample restricted to continuously innovative manufacturing firms (2% of firms; 98% of R&D); conclusions may not generalise.
2. No transitional dynamics between stationary equilibria; welfare gains exclude adjustment costs.
3. Closed economy; no trade, multinationals, or international knowledge flows.
4. Skilled labour relocates instantaneously; in reality R&D labour is highly specialised and slow to reallocate.
5. Two-type approximation; a richer type space would allow more nuanced policy targeting.

11.14 Policy Implications

- **R&D subsidies are relatively ineffective.** +0.63% welfare for 1% GDP cost — less than half the gain from an operation tax.
- **Taxing incumbent operations is powerful.** A 69% tax on fixed costs (8% of revenues on average) raises growth from 2.26% to 2.54% and welfare by 1.4%.
- **Operation subsidies are harmful.** They increase low-type product line shares and depress R&D; welfare falls.
- **Industrial policy should discriminate by exit margin, not by innovation rate.** Policies that target the exit margin achieve positive selection effects that R&D subsidies cannot.

11.15 Overall Assessment

A technically demanding, ambitious paper that marries endogenous growth theory, structural estimation, and policy analysis in a unified framework. The central result—operation taxes dominate R&D subsidies—is counterintuitive, well-motivated, and robustly derived. The model is tractable (franchise values in closed form), well-estimated (18 moments, strong non-targeted fit), and policy analysis is comprehensive.

Quality: Excellent. A major contribution to endogenous growth, firm dynamics, and policy analysis.

11.16 Relevance to My Research

Directly relevant to Canadian RDC research on industrial policy effectiveness. The structural SMM approach is applicable to Canadian Innovation Survey data linked to LEAP/T2 corporate tax records. The central question—whether SR&ED tax credits disproportionately benefit low-type vs. high-type Canadian firms—would be a natural application with first-order policy implications.

11.17 Potential Research Extensions

- Apply the Acemoglu et al. framework to Canadian data (LEAP/T2/ Innovation Survey); test whether SR&ED credits disproportionately benefit low-type firms

as predicted.

- Extend to a small open economy to capture import competition as a mechanism for forcing low-type firm exit (analogous to the operation tax).

11.18 Research Gaps

- The political economy of why operation taxes are not observed in practice, despite their welfare superiority, is unmodelled.
- No analysis of heterogeneity in policy effects across industries with different innovation intensities (high-tech vs. low-tech).

11.19 Methodological Lessons

1. The franchise value additivity result (Lemma 1) reduces the state space from the full product portfolio to a scalar per product line, enabling closed-form solutions even in a complex GE environment.
2. SMM with 18 moments and 8 parameters is overidentified; matching non-targeted moments provides external validation that goes beyond the targeted moments.
3. The policy comparison framework—all interventions costing exactly 1% of GDP—enables clean welfare comparison across fundamentally different policy instruments.

Five-Sentence Summary

Acemoglu et al. (2018) develop a quality-ladder endogenous growth model featuring high- and low-type firms (differing in innovative capacity) with endogenous exit, creative destruction, and type transitions, estimated by SMM on 18 moments from U.S. Census Bureau microdata covering 98% of U.S. industrial R&D. The model's key market failure is the misallocation of skilled labour between low-type firm operations (generating little social value) and R&D by high-type firms (generating large positive spillovers), leading to underinvestment in socially valuable innovation. Policy analysis shows that incumbent R&D subsidies are relatively ineffective (+0.63% welfare for 1% of GDP) because they benefit both firm types indiscriminately, while a 69% operation tax generates a strong positive selection effect (+1.40% welfare) by disproportionately inducing exit among low-type firms and freeing skilled labour for high-type R&D. The social planner, controlling type-specific exit thresholds and R&D rates, achieves +4.47% welfare and raises growth from 2.26% to 2.94% by dramatically raising the low-type exit threshold and redirecting skilled labour to high-type R&D. The central policy lesson is that the standard tool of industrial policy—R&D subsidies—is not merely ineffective but is dominated by policies that accelerate exit of low-productivity incumbents.

Quick Reference

Key contribution	Incumbent operation taxes dominate R&D subsidies for welfare: taxes fall disproportionately on low-type firms near the exit margin, freeing skilled labour for R&D by high-type firms with a 1.4% welfare gain vs. 0.6% from R&D subsidies.
Key weakness	Sample restricted to continuously innovative manufacturing firms (2% of all firms); no transitional dynamics; closed economy; instantaneous skilled labour reallocation.
Most interesting gap	The political economy of why operation taxes are not observed in practice despite their welfare superiority—incumbents lobby against exit-inducing policies even when they reduce aggregate growth.
Publishable extension	Apply the framework to Canadian data (LEAP/T2/Innovation Survey); test whether SR&ED credits disproportionately benefit low-type vs. high-type Canadian firms, and estimate the implied welfare cost of the current design.

Paper 12

Hopenhayn, Neira & Singhania (2022)

From Population Growth to Firm Demographics: Implications for Co

Authors: Hugo Hopenhayn (UCLA), Julian Neira (Exeter), Rish Singhania (Exeter)

Journal: *Econometrica* 90(4): 1879–1914

Year: 2022 **Field:** Macroeconomics; Firm dynamics; Labour economics

Classification: Structural (GE model + quantitative calibration + historical decomposition)

12.1 Research Question

Can changes in population/labour force growth—primarily driven by the baby boom and subsequent demographic slowdown—provide a unified quantitative explanation for three major secular trends in the U.S. economy: (i) rising concentration of employment in large firms; (ii) declining firm entry rates; and (iii) declining aggregate labour share?

12.2 Motivation

Three well-documented but apparently disconnected secular trends in the U.S. economy since the late 1970s have attracted separate literatures:

- **Rising concentration:** employment share in firms with 250+ employees rose from 51.6% to 57.3% (1978–2014).
- **Declining dynamism:** U.S. firm entry rate fell by approximately 6 percentage points.
- **Declining labour share:** aggregate share of GDP going to labour has fallen substantially.

The paper proposes a single unified mechanism: the baby boom increased firm entry rates in the 1970s, creating a cohort of “baby boom firms”; as these firms aged, the firm-age distribution shifted toward older, larger, less dynamic firms—mechanically generating all three trends simultaneously through a simple accounting identity.

12.3 Main Contribution

1. **Empirical documentation:** Using BDS data (1978–2014), shows that once firm-age controls are included, the time trends in concentration, exit rates, and average size *reverse sign*—the aggregate trends are entirely driven by changes

in the firm-age distribution.

2. **Invariance theorem (Corollary 1):** In a general class of GE models (perfect competition, CES monopolistic competition, variable markup oligopoly), equilibrium allocations *conditional on firm age are invariant to changes in population growth*. All aggregate changes must therefore come from changes in the firm-age distribution.
3. **Dynamic entry equation (Corollary 2):** Characterises firm entry as a history-dependent distributed-lag equation, enabling quantitative simulation from historical labour force data.
4. **Quantitative analysis:** Feeds historical U.S. labour force data (1940–2014) through the model; replicates the observed entry rate decline, concentration, average size, exit rates, and the aggregate labour share (including the baby-boom spike and subsequent decline).

12.4 Relationship to the Literature

Synthesises three previously disconnected literatures: (i) business dynamism (Haltiwanger, Jarmin, Miranda 2011; Decker et al. 2014, 2020); (ii) rising concentration (Autor et al. 2020; Grullon et al. 2019); (iii) labour share decline (Karabarbounis–Neiman 2014; Kehrig–Vincent 2021). The closest predecessor is Karahan, Pugsley, Şahin (2018), who link population growth to entry rates in a Hopenhayn (1992)-style model. The present paper extends this by providing conditions applicable to imperfect competition, characterising transitional dynamics explicitly, and showing that baby-boom transitional effects explain 55% of the entry rate decline — more than the long-run effect (45%).

12.5 Theoretical Framework and Economic Mechanism

Accounting Identity

The fundamental identity of the paper:

$$\lambda_t = \frac{\dot{N}_t}{N_{t-1}} \frac{e_{t-1}}{e_t} - 1 + \xi_t, \quad (12.1)$$

where λ_t is the firm entry rate, $\dot{N}/N$ is labour force growth, $e_t = N_t/M_t$ is average employment per firm, and ξ_t is the aggregate exit rate. Entry rate equals labour force growth minus growth in average size plus exit rate.

General Framework

Labour endowment N_t inelastically supplied. Firm profits $\pi(s, z)$ and employment $n(s, z)$ are increasing in idiosyncratic state s and aggregate state z , under a Markov process $F(s'|s)$. The firm’s Bellman equation:

$$v(s, z_t) = \max\{0, \pi(s, z_t) + \beta \mathbb{E}[v(s', z_{t+1})|s]\}, \quad (12.2)$$

with exit threshold $s_t^* = \inf\{s \mid \pi(s, z_t) + \beta \mathbb{E}[v(s', z_{t+1})|s] > 0\}$. Free entry pins down z^* via $v^e(z^*) = \int v(s, z^*) dG(s) - c_e = 0$.

Invariance of Age Profiles (Corollary 1)

Because the aggregate state z^* is determined solely by the zero-profit condition for entrants—*independent of N_t* —adjustments in aggregate quantities occur entirely through changes in the mass of entrants m_t , leaving age profiles unchanged:

Exit rates by age $\tilde{\xi}_a$, Average firm size by age $\tilde{e}_a$, Size distribution by age
are all invariant to changes in N_t .

This result holds for all models satisfying the general framework (three examples given: perfect competition, CES monopolistic competition, variable markup oligopoly).

Dynamic Entry Equation (Corollary 2)

The invariance of age profiles implies that market clearing $N_t = m_t(S_0\tilde{e}_0 + c_e) + \sum_{a=1}^t m_{t-a}S_a\tilde{e}_a$ uniquely determines:

$$m_t = \frac{N_t - \sum_{a=1}^t m_{t-a}S_a\tilde{e}_a}{S_0\tilde{e}_0 + c_e}, \quad (12.3)$$

where S_a is the probability of surviving to age a and $\tilde{e}_a$ is average firm size at age a . Entry is *history-dependent*: current entry depends on the entire distributed lag of past entry masses $\{m_{t-a}\}$.

Long-Run Multiplier (Proposition 2)

When $\tilde{\xi}_a$ is decreasing in a (older firms have lower exit rates, as in the data), the long-run aggregate exit rate ξ^{SS} is *increasing* in population growth g . Therefore, the effect of population growth on the entry rate is amplified:

$$\lambda^{SS} = g + \xi^{SS} \Rightarrow \frac{d\lambda^{SS}}{dg} = 1 + \frac{d\xi^{SS}}{dg} \approx 1.5 \quad (12.4)$$

for the U.S. economy. A 2 pp decline in labour force growth generates ≈ 3 pp decline in the long-run entry rate (long-run effect alone).

Baby Boom Transitional Dynamics

The baby boom raised firm entry rates in the 1960s–70s, creating a glut of firms. Three reinforcing feedback loops amplify the direct effect of declining population growth:

1. **Exit rate feedback:** Aging distribution lowers aggregate exit rate ξ_t (older firms exit less) $\Rightarrow$ entry falls further.
2. **Average size feedback:** Aging distribution raises average firm size e_t (older firms are larger) $\Rightarrow$ entry falls further.
3. **Baby boom glut:** The large cohort of baby-boom firms has not yet died off; as of 2014, the firm-age distribution is older than the new steady state would imply.

12.6 Data

- **Business Dynamics Statistics (BDS):** U.S. Census Bureau, 1978–2014; near-universal coverage of private-sector firms with paid employees; entry/exit rates, average firm size, concentration, and job flows by firm age and sector.
- **Historical labour force data:** 1940–2014 (BLS civilian labour force; Survey of Current Business entry rates 1940–1962).
- **Structural calibration:** GMM targeting 1978–1983 moments: entry rate (13.14%), average entrant size (5.96), concentration of entrants, 5-year firm growth (73.86%), 5-year exit rate (55.01%), large-firm employment shares.
- **Labour share:** Karabarbounis–Neiman (2014); Koh, Santaaulàlia-Llopis, Zheng (2020).

12.7 Empirical Strategy

Three complementary approaches:

1. **Regression evidence (Table I):** Regress exit rate, average firm size, and concentration on year trend, with and without firm-age controls. The reversal of sign when age controls are included is the core identification.
2. **Shift-share accounting:** Compute

$$\text{Change if Aging Only} = \frac{(\text{Unconditional trend} - \text{Age-conditional trend}) \times 36}{\text{Level}_{1978}}$$

to isolate the contribution of the changing age distribution.

3. **Dynamic entry equation simulation:** Estimate the firm employment stochastic process by GMM; feed historical labour force data from 1940 into equation (12.3) and iterate forward to generate aggregate time series.

12.8 Identification Strategy

The key identifying assumption is the **invariance of age profiles**: exit rates, average size, and size distributions by firm age do not change with population growth. This is directly validated by the regression evidence:

- After controlling for firm age, the exit rate trend becomes slightly *positive* (consistent with no within-age-group change).
- After controlling for firm age, average size and concentration trends become *negative* (consistent with mild within-age declines).

The aggregate positive trends in size and concentration are entirely accounted for by the shifting firm-age distribution.

Important caveat: The invariance result requires a perfectly elastic supply of potential entrants proportional to population. If the pool of potential entrepreneurs is fixed, the mechanism is weaker.

12.9 Main Results

Regression Evidence (Table I)

Variable	Aggregate trend	Controlling for age	Sign reversal?
Exit rate	−0.043/yr ***	+0.011/yr	Yes
Average firm size	+0.096/yr ***	−0.143/yr ***	Yes
Concentration	+0.159/yr ***	−0.069/yr ***	Yes

Decomposition of 6 pp Entry Rate Decline (Table V)

Component	Magnitude	Share
Long-run effect (steady state)	−2.76 pp	45%
1978 transition effect (baby boom entry)	−1.47 pp	24%
2014 transition effect (baby boom aging)	−1.84 pp	30%
Residual	−0.01 pp	1%
Total	−6.08 pp	100%

Of the long-run effect: direct labour force growth decline accounts for 1.88 pp; long-run multiplier effect on exit rates adds 0.88 pp.

Model Fit for Aggregate Variables (Figure 8)

Aging alone (model) generates:

- 16% decline in exit rate (data “aging only”: 18%)
- 41% increase in average firm size (data: 43%)
- 15% increase in concentration (data: 16%)

Labour Share and Job Reallocation

Firm aging generates the hump-shaped post-WWII pattern in the aggregate labour share: rising until ≈ 1980 (baby-boom entrants are small, high-labour-share) and falling after (firms age and grow). Firm aging explains:

- 48% of the decline in the job creation rate
- 38% of the decline in the job destruction rate
- 32% of the decline in the job reallocation rate

12.10 Economic Magnitude and Interpretation

The 6 pp decline in the U.S. firm entry rate represents a massive structural change. The paper shows that baby-boom transitional dynamics (55% of total) are quantitatively larger than the long-run steady-state effect (45%), meaning the timing of

the demographic cycle matters as much as its level. With labour force projections through 2060, entry rates are projected to partially recover by ≈ 2030 as the baby boom cohort exits and millennial births enter the labour force.

12.11 Mechanisms

Three reinforcing feedback loops amplify the direct demographic effect:

1. **Exit rate feedback:** Declining entry $\rightarrow$ aging distribution $\rightarrow$ lower aggregate exit rate (older firms exit less) $\rightarrow$ even lower entry.
2. **Average size feedback:** Aging distribution $\rightarrow$ higher average firm size (older firms are larger) $\rightarrow$ lower entry demand per unit of labour.
3. **Baby boom glut:** The large cohort of firms created in the 1970s continues to suppress entry in 2014 because the 2014 firm-age distribution is older than the new steady state would predict.

12.12 Robustness Checks

- Results hold in all 7 major BDS sectors (Table II); aging is quantitatively important across every sector.
- Sector controls do not remove aggregate trends; they deepen them for size and concentration.
- Invariance assumption validated by regression evidence (age controls flip sign of all three trends).
- Shift-share exercise using data levels and model age weights yields similar results.
- Alternative explanations evaluated within the model: higher entry costs, economies of scale, lower diffusion rates—each can explain some facts but is inconsistent with others.
- WW2 fluctuations: model generates entry rate fluctuations matching Survey of Current Business data for 1940–1962.

12.13 Limitations

1. **Perfectly elastic entrant supply:** Requires the number of potential entrants proportional to population; weakened if the entrepreneurial pool is fixed.
2. **Constant aggregate state:** Invariance result requires risk-neutral households or small open economy; CRRA with endogenous interest rates may partially offset the mechanism.
3. **No welfare analysis:** The paper does not characterise whether the shift to older, larger firms is efficient or inefficient.
4. **Labour share mechanism:** Requires overhead labour (fixed cost) to generate the size–labour-share relationship; maintained assumption rather than derived from micro-foundations.

12.14 Policy Implications

- **Declining entrepreneurship is a demographic phenomenon.** Policies targeting startup subsidies address the symptom; the root cause (declining birth rates) will not be reversed by business policy.
- **Concentration is not necessarily a competition problem.** Rising concentration reflects a shift in the age distribution of firms, not necessarily market power. This complicates antitrust interpretations.
- **Labour share decline is partly demographic.** Interventions targeting labour market institutions or trade may address only a subset of the decline.
- **Job reallocation decline.** 32–48% of the decline in gross job reallocation is explained by firm aging; restoring dynamism requires addressing the demographic slowdown, not just regulatory reform.

12.15 Overall Assessment

Hopenhayn, Neira, and Singhanian (2022) achieve something rare in macroeconomics: a parsimonious unified explanation for three major trends that have separately occupied large literatures. The invariance result, applicable across a general class of GE models, is a rigorous theoretical contribution. The transition dynamics analysis—distinguishing long-run and baby-boom transitional effects—is both analytically and quantitatively careful. The empirical evidence (sign reversal with age controls) is compelling.

Quality: Outstanding. A landmark paper in firm dynamics and macroeconomics.

12.16 Relevance to My Research

Highly relevant for RDC research on Canadian firm dynamics. Canada has experienced similar demographic trends. The Business Register/LEAP data could test the HNS mechanism for Canada: does firm aging explain declining Canadian entry rates? The heterogeneous provincial demographics (Quebec birth rate fluctuations, Alberta immigration-driven growth) could provide within-country identifying variation.

12.17 Potential Research Extensions

- Test the HNS mechanism for Canada using LEAP data; exploit provincial variation in demographic patterns (Quebec vs. Alberta vs. Ontario) as quasi-experimental variation in the key driving force.
- Extend the model to incorporate immigration separately from native birth rates; test whether immigration-driven labour force growth has different effects on firm demographics than birth-rate-driven growth.

12.18 Research Gaps

- The welfare consequences of the baby-boom firm aging are uncharacterised: is the shift toward older, larger firms efficient (positive selection) or inefficient (protection of incumbents from younger challengers)?

- The model does not account for the global nature of the trends; similar patterns in European economies with different demographics suggest additional forces beyond demographics alone.

12.19 Methodological Lessons

1. The accounting identity (12.1) organises all subsequent analysis without requiring a full structural model; it enables clear decomposition of what drives entry rate changes.
2. The regression reversal (age controls flip all three trend signs) is a simple, convincing empirical strategy for establishing the proximate mechanism, regardless of the underlying model.
3. The invariance result reduces a complex GE problem to a demographic accounting exercise, making the dynamic entry equation (12.3) applicable across market structures.
4. Separating transitional from long-run effects reveals that one-half of the observed trend is transitory (baby boom)—the economy is not in steady state and may partially recover, with major implications for policy urgency.

Five-Sentence Summary

Hopenhayn, Neira, and Singhanian (2022) document that three major secular trends in the U.S. economy—rising employment concentration, declining firm entry rates (−6 pp from 1978–2014), and declining labour share—all reverse sign once firm-age controls are included in regressions, establishing that they are driven entirely by changes in the firm-age distribution rather than within-age-group trends. They develop a general theoretical framework, applicable to perfect competition, CES monopolistic competition, and variable markup oligopoly, showing that equilibrium allocations conditional on firm age are invariant to changes in population growth, and derive a dynamic entry equation linking current entry to the entire distributed lag of past entry via survival probabilities and average firm size by age. Feeding historical U.S. labour force data (1940–2014) into this equation, the model replicates the 6 pp decline in the entry rate—with 45% attributable to the long-run multiplier effect of declining labour force growth (elasticity ≈ 1.5) and 55% to baby-boom transitional dynamics (aging of firms founded in the 1970s). Firm aging also generates the hump-shaped post-WWII pattern in the aggregate labour share and accounts for 32–48% of the decline in gross job reallocation rates. The paper provides the first unified quantitative explanation linking baby-boom demographics to declining entrepreneurship, rising concentration, declining labour share, and falling job reallocation—trends previously treated as separate phenomena requiring separate explanations.

Quick Reference

Key contribution	First unified quantitative explanation linking baby-boom demographics to declining entrepreneurship, rising concentration, declining labour share, and falling job reallocation through firm aging; invariance theorem applicable across a general class of GE models.
Key weakness	Requires perfectly elastic entrant supply proportional to population; no welfare analysis of the aging transition; labour share mechanism requires maintained assumption of overhead labour.
Most interesting gap	The welfare consequences of baby-boom firm aging are uncharacterised: is the shift toward older, larger firms efficient (positive selection) or inefficient (protection of incumbents from younger challengers)?
Publishable extension	Test the HNS mechanism for Canada using LEAP data, exploiting provincial variation in demographic patterns (Quebec vs. Alberta vs. Ontario) as quasi-experimental variation; test whether the long-run multiplier elasticity of ≈ 1.5 also holds in Canadian provincial data.